Learning When to Verify Under a Budget: Process Rewards, Representations, and Length Generalization

Raj Kashikar
github.com/raj200501/UltraRL
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Abstract

Reasoning systems must decide when to spend effort checking their own work. We study this decision in an exactly measurable testbed: an agent executes an arithmetic chain that may contain one silent fault, sees a noisy one-bit cue per step, and divides a fixed budget between executing, verifying and repairing steps. We train Double DQN agents with outcome-only and process rewards and several observation encoders, and test them on chains up to 2.3× longer than in training (10 seeds, paired tests).

1. Length generalization depends mainly on the observation encoder. With length-dependent features every reward falls below a never-check baseline on the longest chains; with length-invariant features the reward that pays most for informative checks keeps 0.86 success at lengths 7–9 and 0.74 at 10–14 at selected checkpoints, against 0.94 and 0.90 for an exact planner; at 10–14 a cue-triggered heuristic (0.85) still beats it.
2. Under every reward, agents drift into catching faults without repairing them. This is a learning failure under partial observability, not reward hacking: one observation bit recording that a caught fault is unrepaired removes it under all three rewards, and so does a window of the agent's last 3 actions; a window of 8 fails on long chains.
3. Paying for informative checks makes checking policies appear sooner (median episode 3,500 against 15,000 for outcome-only) and keeps them (24% against 66% of later checkpoints fall back to success ≤ 0.35). A step-correctness bonus, closer to a process reward model, helps less and never resolvably beats outcome-only at selected checkpoints.

These results revise an earlier version of this study, which attributed the length-generalization failure to the reward.

1 Introduction

Reinforcement learning has produced language models that spend test-time compute on long chains of reasoning, including checking and revising their own steps [OpenAI, 2024, DeepSeek-AI, 2025, Kumar et al., 2024], and spending inference compute on search and revision can be worth more than scaling the model [Snell et al., 2024]. Two lines of work make this behaviour a target in its own right. Process supervision rewards individual reasoning steps rather than only final answers [Uesato et al., 2022, Lightman et al., 2024]. Work on chain-of-thought monitorability asks whether reasoning traces stay legible enough to oversee, and warns that optimising against a monitor can make them less so [Baker et al., 2025, Korbak et al., 2025, OpenAI, 2025]. Both lines touch a basic metareasoning question [Russell

and Wefald, 1991, Hay et al., 2012]: with a finite budget, *which* steps are worth checking, and does a learned answer to that question survive when the problem gets longer?

We borrow the term "monitorability-aware" from the original version of this study for a reward that pays for checks which keep a trajectory locally diagnosable. The testbed has no monitor and no notion of trace legibility, so this reward is a toy proxy, and our results bear on process supervision of *checking* rather than on chain-of-thought monitoring.

That question is hard to study in language models, where the location of an error, the value of a check and the faithfulness of a trace [Turpin et al., 2023, Lanham et al., 2023, Chen et al., 2025] are all hard to measure. We therefore isolate it in a small testbed (§3). An agent executes an arithmetic chain that may contain one silent fault. It sees a noisy one-bit cue per step and can execute, verify, undo or commit, each non-commit action consuming a fixed budget. The task admits strong hand-coded policies (a cue-triggered checker solves 85–88% of episodes in every length range we test, and an exact planner that knows the chain length and never goes back to check an earlier step 90–99%), so failures of learning cannot be blamed on an impossible task. Because every quantity is exact, we can separate three things that are entangled in realistic settings: whether an agent learns to check at all, whether it checks the *right* steps, and whether the behaviour transfers to longer chains than it was trained on.

Contributions.

- A small, fast and fully tested environment for budgeted verification, with exact metrics for faults, detection, repair and cue-conditional checking, several reward schemes and observation encoders, hand-coded reference policies, and an exact dynamic-programming planner that gives the best success achievable without going back to check earlier steps, which an exhaustive search shows is the unrestricted optimum on chains of up to 7 steps (§3–4).
- Evidence that length generalization is dominated by the observation encoder. Length-dependent features make every held-out observation novel and drive every reward below a never-check baseline on long chains; a length-invariant encoder recovers most of the performance. A control shows that removing only the position features is not enough (§6.3).
- A failure mode shared by all rewards, where agents detect faults but do not repair them. It is not reward hacking but a consequence of partial observability: one bit recording that a caught fault is unrepaired removes it under every reward, and so does a history of the agent's last 3 actions, while a history of 8 actions fails on long chains (§6.2, §6.9).
- Evidence that paying for informative checks helps agents find checking policies sooner, under every learner setting and observation we tried, and keep them, under every learner setting and with the invariant observation or the repair-pending bit (with an action history neither reward's checkpoints collapse); a step-correctness bonus, the analogue of a process reward model, helps less and plateaus lower (§6.1, §6.8, §6.9). Ablations show that any check payment gives most of the benefit on training lengths (§6.7).
- An evaluation under shifts in fault rate, verification cost, budget and cue quality (the agents degrade when checks cost more or the slack changes), a check of the encoder and reward comparisons under other learner settings, and an analysis of training instability (§6.6–6.8).

A correction. An earlier version of this project, using the same environment and rewards, concluded that monitorability-aware RL "remains brittle under longer held-out chains". A controlled re-run reversed that conclusion. The main cause was the length-dependent features of the original observation encoder, compounded by 4× less training and last-checkpoint evaluation. The earlier pipeline also used a localization metric that gave credit for faults nobody checked, and it reused training seeds for one seed's evaluation episodes (harmless here, but a protocol error). Appendix I lists every change. We report this because, at larger

scale, the same confounds (length-dependent features and unstable policies) can make a reward look responsible for failures it did not cause.

2 Related Work

Process supervision and verification. Step-level feedback improves the reliability of mathematical reasoning [Uesato et al., 2022, Lightman et al., 2024], and RL on verifiable tasks can elicit self-checking [DeepSeek-AI, 2025, Xie et al., 2025]. These works ask whether process feedback helps. We ask a narrower question in a controlled setting: when a process reward helps an agent learn to check, what has it learned, and does that transfer? Our process rewards are hand-specified with privileged access to step correctness, so they stand in for an *accurate* process reward model. Process verifiers can also be trained automatically from rollouts [Wang et al., 2024], and rewarding a step's progress toward a correct answer, rather than its correctness, gives a better RL signal [Setlur et al., 2025]. Our finding that paying at the repair step stabilises training (§6.9) is in the same spirit: what a process reward pays for, and when, matters.

Reward misspecification. Agents can score well on a specified reward while failing at the intended task [Krakovna et al., 2020, Skalse et al., 2022, Pan et al., 2022], and process rewards used during RL for language-model reasoning can be exploited in this way [Gao et al., 2024], for instance by repeating steps that earn a step reward. Our largest step bonus has that loophole; paying it once per step closes it without removing its gain over outcome-only (§6.1). The detect-without-repair failure we find (§6.2) looks superficially like such gaming. However, it scores *worse* on the specified reward than the intended behaviour, so it is a failure of optimisation rather than of specification. It is loosely the reverse of a difficulty reported for language models, which struggle to correct their reasoning without external feedback [Huang et al., 2024] and often fail to find their own errors but can correct them once the error is located [Tyen et al., 2024]: our agents locate faults reliably and then fail to repair them, although repair here is a fixed two-action sequence rather than a reasoning problem.

Monitorability and faithfulness. Reasoning traces can be informative about model behaviour [Baker et al., 2025, Korbak et al., 2025], but they are not guaranteed to be faithful [Turpin et al., 2023, Lanham et al., 2023, Chen et al., 2025], and optimisation pressure on traces can induce obfuscation [Baker et al., 2025]. Our "monitorability-aware" reward is a toy instance of rewarding traces that remain locally diagnosable: it pays for checks that catch or rule out faults and penalises long unchecked stretches. Our environment has no notion of a model hiding its reasoning, so we make no claims about faithfulness.

Metareasoning and adaptive computation. Deciding which computation to perform next is classical metareasoning [Russell and Wefald, 1991, Hay et al., 2012], and adaptive computation time learns how long to think [Graves, 2016]. Learning which computations are worth performing has also been studied directly [Callaway et al., 2018]. In RL terms, a check is a costly, informative action under partial observability, as in work on RL with observation costs [Bellinger et al., 2021, Nam et al., 2021]. Our agents learn cue-conditional checking from reward alone; they are not given, and with the invariant encoder cannot represent, the posterior over fault locations.

Generalization and evaluation in RL. Deep RL agents overfit their training distribution [Cobbe et al., 2019, Kirk et al., 2023], and neural networks often fail to generalize to longer inputs than seen in training [Anil et al., 2022, Delétang et al., 2023]. Our length-generalization result is a small, controlled instance of this: in our runs it tracks whether observation features change meaning with length or take values, or combinations of values, that training-length chains never produce (§7). Value-based learning can converge to poor values when distinct states share an observation [Whitehead and Ballard, 1991, Singh et al., 1994, McCallum, 1996], and recurrent Q-networks are the standard remedy [Hausknecht and Stone, 2015]; such aliasing turns out to drive the detect-without-repair failure we find (§6.2, §6.9). Methodologically we follow recommendations to report many seeds with interval

estimates rather than point estimates [Henderson et al., 2018, Agarwal et al., 2021]. We also document policy churn [Schaul et al., 2022]: greedy policies that change abruptly between updates, which makes last-checkpoint evaluation unreliable.

3 A Testbed for Learning When to Verify

Task. Each episode is an arithmetic chain $s_0 \to s_1 \to \cdots \to s_d$ of depth d: the start value $s_0 \in \{2, \ldots, 6\}$ is followed by d randomly sampled transitions (add, subtract, or multiply by a small constant). With probability $p_f = 0.8$, exactly one transition h is *faulty*: until the fault has been located, executing it applies a corrupted operation, so the agent's running value silently diverges from the true value. Conditional on a fault, h is drawn from a prior that peaks in the middle of the chain. Each transition carries a one-bit *cue* c_t that fires with probability 0.85 on the faulty transition and with probability 0.15 on every other transition. The cue plays the role of a cheap, noisy signal that a reasoning step looks suspicious.

Actions and budget. The agent chooses among four actions. `THINK` executes the next transition. `VERIFY` exactly checks the most recently executed transition and returns SAFE or UNSAFE; the check is *local*, so it does not detect corruption inherited from an earlier unchecked fault. The first UNSAFE result locates the fault, after which re-executing it (`REVISE`, which undoes the last step, followed by `THINK`) applies the correct operation. `COMMIT` ends the episode, which succeeds if and only if all d transitions have been executed and the final value is correct. `THINK`, `VERIFY` and `REVISE` each cost one unit of a budget of $d + 6$ units. The agent can therefore afford six actions beyond the chain itself: enough to check some steps and repair one fault, but not to check every step of a long chain.

The environment is deliberately small. All quantities of interest are exact: whether a fault was present, whether and where it was caught, and which steps were checked. The question is not whether the task is hard, but whether reinforcement learning (RL) finds and keeps the right *allocation* of checking effort, and whether that allocation transfers to longer chains.

Observations. The agent never observes the running value, the true value or the fault location. We compare two observation encoders (Table 1). Both expose the local information a hand-coded checking policy needs (cue, budget slack, unchecked gap, last check result). The invariant encoder is a strict reduction of the absolute one: it replaces position and depth by two boundary flags and drops the budget fraction, so the number of remaining steps cannot be recovered from it.

- **absolute** (10-d): normalised position $t/9$, depth $d/9$, budget fraction, normalised budget slack, normalised unchecked gap, cue, a revised-last-step flag and a one-hot of the last verification result. This is the encoding used in the original version of this study.
- **invariant** (9-d): flags for $t = 0$ and $t = d$, budget slack $\text{clip}(b_t - (d - t), -10, 10)/10$, unchecked gap $\min(g_t, 3)/3$, cue, the revised flag and the verification one-hot.

Here b_t is the remaining budget and $g_t = t - \max(\text{checked indices})$ the number of steps executed since the highest-index checked step. Budget slack $b_t - (d - t)$ is the number of non-`THINK` actions still affordable. On held-out lengths the absolute encoding produces feature values never seen in training (for example $d/9 > 6/9$); the invariant encoding rarely does (§6.3). Neither encoding gives the agent memory of earlier UNSAFE results. As a consequence, a genuine repair (UNSAFE, `REVISE`, `THINK`) and a no-op `REVISE`+`THINK` after a SAFE check lead to identical observations (§6.9 adds a one-bit memory).

4 Rewards, Learner and Reference Policies

Base reward. All agents receive $+1$ for a successful commit, -1.2 for a failed one, and -0.04 per unit of budget spent.

Process rewards. We compare three reward schemes, keeping the coefficients of the earlier version:

Table 1: Observation features. The invariant encoder replaces absolute position and length by boundary flags, so remaining length cannot be recovered from it.

Information	Absolute encoder	Invariant encoder
Position	$t/9$, depth $d/9$	$[t{=}0]$, $[t{=}d]$
Budget	b_t/B, $\mathrm{clip}(0.5 + \mathrm{slack}/2B, 0, 1)$	$\mathrm{clip}(\mathrm{slack}, \pm 10)/10$
Unchecked gap	$g_t/9$	$\min(g_t, 3)/3$
Cue, revised flag, last check	shared	shared

- **Outcome-only**: the base reward alone.
- **Naive process**: +0.015 for each `THINK` that leaves the running value correct, and +0.02 for each first-time informative check. These are the kind of dense "step is correct" signals a process reward model provides [Uesato et al., 2022, Lightman et al., 2024].
- **Monitorability-aware** (the earlier version's name, kept for continuity): rewards checks that make the trajectory more locally diagnosable. It is a toy proxy only. The testbed has no monitor and no notion of trace legibility, so results about this reward say nothing directly about chain-of-thought monitorability:

$$r_t^{\mathrm{MA}} = 0.02\,\mathbf{1}[\text{safe check}] + 0.25\,\mathbf{1}[\text{fault caught}] + 0.18\,\mathbf{1}[\text{informative check, } c_t{=}1] - 0.02\,\mathbf{1}[\text{safe check, } c_t{=}0] - 0.06\,\mathbf{1}[\text{wasted check}] - 0.3\,\frac{g_{t+1} - g_t}{d}. \tag{1}$$

  A check is *informative* if it is the first check of that step and either confirms a correct value or catches the fault. It is *wasted* if it repeats a check, or checks a step whose running value is already wrong, that is, a step downstream of an unrepaired fault, whether or not that fault was caught.

Two properties of this reward, inherited unchanged from the earlier version, matter for what follows. First, the fault bonus is paid when the fault is *caught*, whether or not it is then repaired. Second, the gap term telescopes to $-0.3\,g_T/d$ over an episode. It is therefore close to potential-based shaping [Ng et al., 1999] plus a terminal penalty that depends only on how far the last *checked index* is from the end. One check of the final step cancels it, and any action that lowers g is refunded, including a `REVISE` or a check that is otherwise "wasted". For a check downstream of a hidden fault, the refund often exceeds the 0.06 waste penalty. The non-gap terms are not potential-based and can change the optimal policy. Both process rewards use privileged information (the hidden correctness of the running value), standing in for an accurate process reward model.

Learner. Every learned agent is a masked Double DQN [Mnih et al., 2015, van Hasselt et al., 2016] with a 2×64 ReLU MLP (5,060 parameters with the invariant observation; 6,020 and 7,620 for the 3- and 8-action history encoders of §6.9), Huber loss, Adam (learning rate 10^{-3}), a 25k-transition uniform replay buffer, batch size 64, a hard target update every 200 environment steps, and ϵ-greedy exploration annealed linearly from 0.5 to 0.02 over the first 85% of training. Unavailable actions are masked both when acting and in the bootstrap target. Each run trains for 20,000 episodes on chains of length 3–6. These are the earlier version's hyperparameters, unchanged except for 4× more training. Before the main experiment, a two-seed pilot tried six settings for the monitorability-aware agents (two learning rates, two target-update periods, and Polyak averaging at two learning rates) and Polyak averaging for outcome-only agents (Appendix B, Table 9; archived in `results/pilot/`). The original setting had the highest training-length success of the four learning-rate and target-period combinations for the invariant agents and was close to the best for the absolute ones; Polyak averaging did about as well or better. We kept the original setting for continuity with the earlier version. The pilot is weak evidence: with two seeds and an earlier revision of the code, its hard-update-every-1,000 result for the invariant agents did not reproduce in the five-seed sensitivity study (§6.8). The pilot also measured success at lengths 7–9, so those

lengths were not completely untouched by this choice; lengths 10–14 were. §6.8 re-runs the encoder and reward comparisons under three of the alternatives with five seeds each.

Reference policies. We also evaluate four hand-coded policies that see only observable quantities. *Always THINK* never checks. *Verify every step* checks each new step. *Cue-triggered* checks a new step if and only if its cue fired. *Budget-aware* checks cued steps whenever a repair stays affordable, and also checks un-cued steps while the budget slack exceeds that requirement by a margin m. We pick m=1, the value that maximises success on training lengths (`scripts/tune_heuristic.py`). All four policies repair immediately after an UNSAFE check.

Exact planner. As a strong reference we compute the success-maximising policy exactly by dynamic programming (`src/ultrarl/optimal.py`). It considers policies that decide whether to check each step right after executing it; going back to check an earlier step costs at least two extra units. Under this restriction the problem has a small sufficient statistic: if an executed step held the fault unchecked the episode is lost, so only the event that every executed step is clean matters, and on that event the posterior over the fault is the prior restricted to the remaining steps. The value then depends only on the number of executed steps and the remaining slack. An exhaustive search over every history-dependent policy, including ones that undo steps to check earlier ones (`scripts/check_planner.py`), finds the same value for every evaluation environment at lengths 2–7, so the restriction does not bind there. The planner knows the chain length and the environment's parameters, including shifted ones, so it is a reference for what is achievable rather than a fair baseline; tests check that its policy achieves the planned success in simulation and that it matches the exhaustive search on short chains.

5 Evaluation Protocol

Seeds and episodes. Every learned condition is trained with 10 independent seeds. Each policy is evaluated on 500 episodes per seed and setting. Training, learning-curve validation and final evaluation draw episodes from independent random-number streams. Policies observe only depth, position, budget, cue and check results, so on training lengths the same small set of task configurations recurs in all three streams; training-length evaluation is therefore in-distribution by design, and held-out lengths are the only genuinely unseen configurations. Methods are evaluated on identical episodes for a given seed, so comparisons are paired.

Settings. We report success on training lengths (3–6), on held-out lengths 7–9, and on lengths 10–14. On 7–9 we additionally shift the fault rate ($p_f \in \{0.6, 0.9\}$), the verification cost ($c_V \in \{2, 3\}$), the budget slack (4 or 8) and the cue quality (clean: 0.95/0.05; noisy: 0.60/0.35), without retraining.

Checkpoint selection. DQN policies on this task change abruptly during training (§6.8), so the last checkpoint is a noisy sample of what a run has learned. We greedily evaluate every agent on 200 validation episodes of *training-length* chains every 1,000 episodes, and keep the checkpoint with the highest validation success. Held-out lengths are never used for selection; the config loader rejects any selection setting that is not the training distribution. Appendix C repeats the main results and the ablations for the last checkpoint. The encoder gap and the advantage of the monitorability-aware reward over outcome-only hold there too; parity with the cue-triggered heuristic and the ranking of the ablations do not (§6.8, §6.7).

Metrics and statistics. Beyond success we report four diagnostics: *fault detection*, the probability that an UNSAFE check occurs in an episode with a fault; *repair completion*, the probability of success given that the fault was detected; the number of *non-repair* `REVISE`s (those not directly following an UNSAFE check); and *cue-conditional checking*, the probability of choosing `VERIFY` at a decision point whose current step is unchecked, split by its cue c. The last is pooled over decision points, so a step revisited after a `REVISE` counts again. Intervals are 95% Student-t intervals of the mean over seeds. We do not use a percentile

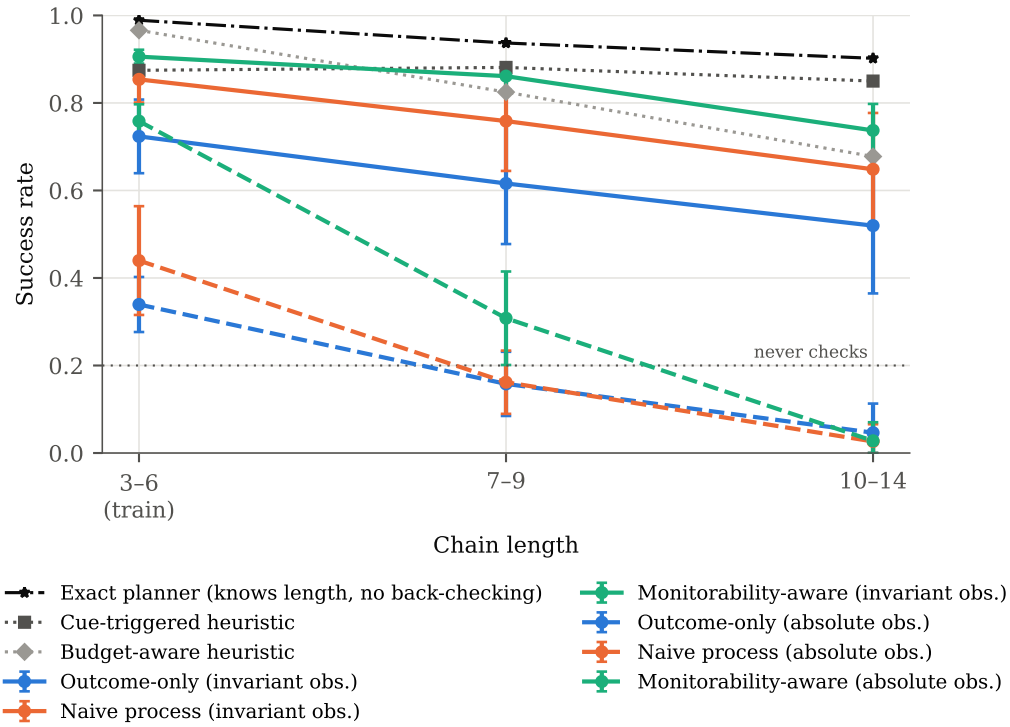


Figure 1: Success on training lengths (3–6) and on longer held-out chains. Colour encodes the reward and line style the observation encoder (solid: invariant, dashed: absolute); error bars are 95% t-intervals over 10 seeds. With absolute features every reward scheme drops below a policy that never checks. With invariant features the monitorability-aware agent is close to the cue-triggered heuristic at lengths 7–9; the exact planner knows the chain length and gives the best success achievable without going back to check earlier steps.

bootstrap, which tends to undercover in samples this small [Efron and Tibshirani, 1993]; Agarwal et al. [2021] find it adequate with 10 runs for aggregates over many tasks, but our comparisons are on a single task. Paired differences between methods report the mean over seeds with a t-interval, the number of seeds favouring each method, and an exact two-sided sign-flip permutation p-value.

6 Results

Table 2 summarises the main experiment and Figure 1 shows the headline result. Unless stated otherwise, numbers are means over 10 seeds with 95% t-intervals, evaluated with the checkpoint selected on training-length validation. Paired differences give the mean difference over seeds, its interval, the number of seeds favouring the first method, and a sign-flip p-value.

6.1 Paying for checks helps agents find checking sooner and keep it

Never checking succeeds exactly when no fault is present, which happens 20% of the time. Any gain above that requires the agent to discover a three-step pattern: check a suspicious step, see UNSAFE, then undo and redo.

Absolute observations. Outcome-only agents rarely acquire this pattern. Their selected checkpoints reach 0.34 [0.28, 0.40] on training lengths, and no seed's validation success ever exceeds 0.5 (0/10). The monitorability-aware reward reaches 0.76 [0.72, 0.80]: a paired difference of +0.42 [0.38, 0.46] in 10/10 seeds (Holm-adjusted $p = 0.012$).

Table 2: Main results (mean over 10 seeds, 95% t-interval). "Detect" is the fraction of faulty episodes in which a check caught the fault; "cost-adj." is success minus 0.05× budget spent. Top block: hand-coded references; bottom block: learned agents (checkpoint selected on training lengths only).

Method	Success 3–6	Success 7–9	Success 10–14	Detect 7–9	Cost-adj. 7–9
Random	0.00 [0.00, 0.01]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.02 [0.01, 0.02]	−0.09 [−0.09, −0.08]
Always THINK	0.20 [0.19, 0.21]	0.20 [0.19, 0.22]	0.20 [0.18, 0.21]	0.00 [0.00, 0.00]	−0.20 [−0.21, −0.18]
Verify every step	0.80 [0.79, 0.81]	0.06 [0.06, 0.07]	0.00 [0.00, 0.00]	0.93 [0.91, 0.94]	−0.63 [−0.64, −0.63]
Cue-triggered	0.87 [0.86, 0.89]	0.88 [0.87, 0.90]	0.85 [0.84, 0.86]	0.85 [0.84, 0.87]	0.33 [0.31, 0.34]
Budget-aware	0.97 [0.96, 0.97]	0.82 [0.81, 0.84]	0.68 [0.66, 0.70]	0.78 [0.76, 0.80]	0.19 [0.17, 0.20]
Planner (no back-checking)	0.99 [0.99, 0.99]	0.94 [0.93, 0.94]	0.90 [0.89, 0.92]	0.92 [0.91, 0.93]	0.33 [0.32, 0.34]
Outcome (inv)	0.72 [0.64, 0.81]	0.62 [0.48, 0.75]	0.52 [0.36, 0.67]	0.61 [0.43, 0.80]	−0.02 [−0.16, 0.12]
Naive process (inv)	0.85 [0.80, 0.91]	0.76 [0.64, 0.87]	0.65 [0.52, 0.78]	0.76 [0.62, 0.90]	0.09 [−0.02, 0.21]
Monitorability (inv)	0.91 [0.89, 0.92]	0.86 [0.83, 0.89]	0.74 [0.68, 0.80]	0.85 [0.83, 0.88]	0.19 [0.15, 0.22]
Outcome (abs)	0.34 [0.28, 0.40]	0.16 [0.08, 0.23]	0.05 [0.00, 0.11]	0.09 [0.03, 0.16]	−0.27 [−0.39, −0.15]
Naive process (abs)	0.44 [0.32, 0.56]	0.16 [0.09, 0.23]	0.03 [0.00, 0.07]	0.30 [0.08, 0.52]	−0.32 [−0.44, −0.21]
Monitorability (abs)	0.76 [0.72, 0.80]	0.31 [0.20, 0.41]	0.03 [0.00, 0.07]	0.60 [0.43, 0.76]	−0.24 [−0.33, −0.15]

Invariant observations. Outcome-only agents *do* find checking policies: some validation checkpoint exceeds 0.5 in 9/10 seeds. They find them late and rarely keep them. The first such checkpoint comes at a median of episode 15,000, against 3,500 for the monitorability-aware reward, and afterwards 66% of their checkpoints have collapsed again, against 24%. Twice the training does not close the gap: with 40,000 episodes, outcome-only agents still collapse in 83% of late checkpoints, and monitorability-aware agents in 22% (5 seeds each, §6.8). After the first 5,000 episodes, 85% [76%, 93%] of their greedy checkpoints are at success ≤ 0.35 (Figure 2). For the monitorability-aware reward the figure is 21% [10%, 32%]; intervals are over seeds. We call these checkpoints *collapsed* throughout. The measure mixes late acquisition with loss: for outcome-only agents, 61% of collapsed late checkpoints come before that seed first exceeds 0.5 (for the monitorability-aware reward, 0%), whereas the retention figures above count only checkpoints after it. Naive process reward pays a token 0.02 for informative checks and 0.015 for every step that leaves the running value correct. It helps partially: its late checkpoints collapse in 62% [51%, 73%] of cases, and fall into a low-detection regime (fault detection below 0.2) in 17% [9%, 26%], against 60% for outcome-only. The monitorability-aware reward pays 0.20–0.43 for an informative check of a cued step and 0.25 for catching an un-cued fault (an un-cued check that finds nothing nets zero from the check terms, apart from the gap-term refund of §4). The two process rewards differ in more than the size of the check payment (the naive reward also pays for correct steps and has no cue, gap or waste terms), so the three rewards should not be read as a dose-response in check payment (a follow-up below separates the naive reward's two terms). The two kinds of collapse also differ. Most collapsed outcome-only checkpoints still check but almost never catch a fault: fault detection is below 0.2 in 60% of all late checkpoints. The ones we inspected (the last checkpoints of five seeds) check once to three times per episode, always the final step and regardless of the cue, so they catch a fault only when it sits in the last step. No collapsed monitorability-aware checkpoint is of this kind (0%); those checkpoints still catch faults but fail to repair them (§6.2).

At selected checkpoints, the monitorability-aware reward beats outcome-only by +0.18 [0.10, 0.26] on training lengths (Holm-adjusted $p = 0.012$). Its advantage over naive process reward is smaller: +0.05 [−0.00, 0.11] ($p = 0.066$) on training lengths and +0.10 ($p = 0.020$) at 7–9.

The process reward that most resembles a learned process reward model, naive process reward, which pays for correct steps, is not reliably better than outcome-only on held-out lengths (§6.3), with the one coefficient setting inherited from the earlier version. What helps most is paying for *checks* (§6.7), and the monitorability-aware reward pays most for the checks a hand-coded cue-triggered policy makes. Its training-length success of 0.91 [0.89, 0.92] lies between that heuristic (0.87) and the tuned budget-aware heuristic (0.97).

Separating and scaling the naive reward. Because the naive reward is the closest analogue of a process reward model, a follow-up sweep (10 seeds each, Table 3) separates its two terms and scales up its step bonus with the check payment removed. The token

Table 3: The naive process reward with its terms separated and its step bonus scaled up (invariant observations, 10 seeds, 95% t-intervals), next to the main experiment's outcome-only and naive agents. "Once": the step bonus is paid only for each step's first correct execution, so that undoing and redoing a correct step earns nothing. "Collapsed": fraction of late checkpoints (after 5,000 episodes) with training-length success ≤ 0.35; "low detection": collapsed with fault detection below 0.2; "late success": mean training-length success of the late checkpoints; "wasted REVISE": non-repair `REVISE`s per episode at lengths 10–14, last checkpoints. Bottom: success at selected checkpoints and, for 7–9, at the last checkpoint.

Reward	Per correct step	Per informative check	Collapsed	low detection	Late success	Wasted REVISE
Outcome-only	–	–	85%	60%	0.32	1.20
Naive process (main experiment)	0.015	0.02	62%	17%	0.43	0.86
Check payment 0.02 only	0	0.02	87%	49%	0.29	0.93
Step bonus 0.015 only	0.015	0	67%	33%	0.39	1.18
Step bonus 0.05 only	0.05	0	51%	20%	0.46	1.38
Step bonus 0.1 only	0.1	0	28%	8%	0.53	1.64
Step bonus 0.05, once per step	0.05 (once)	0	53%	12%	0.46	1.30
Step bonus 0.1, once per step	0.1 (once)	0	36%	8%	0.52	0.85

Reward	Success 3–6	Success 7–9	Success 10–14	Last ckpt. 7–9
Outcome-only	0.72 [0.64, 0.81]	0.62 [0.48, 0.75]	0.52 [0.36, 0.67]	0.33 [0.21, 0.46]
Naive process (main experiment)	0.85 [0.80, 0.91]	0.76 [0.64, 0.87]	0.65 [0.52, 0.78]	0.36 [0.21, 0.51]
Check payment 0.02 only	0.66 [0.50, 0.82]	0.48 [0.32, 0.63]	0.39 [0.26, 0.52]	0.27 [0.16, 0.37]
Step bonus 0.015 only	0.76 [0.64, 0.88]	0.65 [0.55, 0.74]	0.53 [0.41, 0.65]	0.33 [0.17, 0.48]
Step bonus 0.05 only	0.80 [0.77, 0.84]	0.69 [0.62, 0.77]	0.59 [0.49, 0.70]	0.41 [0.28, 0.53]
Step bonus 0.1 only	0.76 [0.75, 0.77]	0.67 [0.58, 0.75]	0.58 [0.48, 0.68]	0.42 [0.31, 0.54]
Step bonus 0.05, once per step	0.82 [0.78, 0.86]	0.73 [0.66, 0.79]	0.61 [0.52, 0.70]	0.42 [0.30, 0.55]
Step bonus 0.1, once per step	0.80 [0.76, 0.83]	0.72 [0.65, 0.79]	0.61 [0.52, 0.70]	0.45 [0.28, 0.61]

check payment alone does nothing measurable: 87% of late checkpoints collapse, against 85% for outcome-only, and its selected checkpoints are 0.19 below the combined reward's on training lengths ($p = 0.006$). The step-correctness bonus alone accounts for most of the naive reward's effect on stability, and more of it helps more. Collapse falls from 67% of late checkpoints at 0.015 per correct step to 51% at 0.05 and 28% at 0.1, and mean late training-length success rises from 0.32 for outcome-only to 0.39, 0.46 and 0.53 (at 0.05 and 0.1 ahead of outcome-only in 10/10 and 10/10 seeds, $p = 0.002$); the trend had not levelled off at 0.1. A dense reward for correct steps, which resumes once a fault is repaired, thus does part of what check payments do.

The bonus as specified is paid for every correct `THINK`, including the re-execution after undoing a correct step. Above twice the per-action cost (0.08), undoing and redoing a correct step therefore has positive reward, the step-repetition exploit of process rewards [Gao et al., 2024], and the 0.1 setting crosses that line. We see no sign that the agents learned to exploit it: on training lengths their last checkpoints waste no more `REVISE`s than outcome-only agents' (1.18 against 1.14 per episode, $p = 0.861$). At 10–14 they waste more than with the smallest bonus (1.64 against 1.18, unresolved, $p = 0.096$). Paying the bonus only for each step's first correct execution, which still pays for repairs, cuts that (0.85, $p = 0.004$) but leaves the stability gain: late training-length success 0.52 against 0.53 ($p = 0.869$) and collapse 36% against 28% ($p = 0.102$), with the same picture at 0.05. The gain over outcome-only does not come from the exploitable loop.

The step bonus stays short of the substantial check payment, though not on every measure. At 0.1 it prevents collapse about as well (28% against 21%, an unresolved difference, $p = 0.406$; paid once per step it collapses more often than with the check payment, 36%, $p = 0.047$) and finds checking about as fast (median first episode above 0.5: 3,000 against 3,500). But its checkpoints plateau lower: mean late success is 0.53 against 0.63 ($p = 0.014$), only 2% of its late checkpoints reach training-length success 0.8, against 39%, and the monitorability-aware agents' selected checkpoints are ahead of every variant of the sweep at every length, by at least 0.09 (resolved in 17 of the 18 comparisons). Against outcome-only, no variant raises selected-checkpoint success resolvably at any length; the largest difference is +0.11 [−0.04, 0.26]. Whether the token check payment adds to the step bonus at selected checkpoints is not resolved (the step bonus alone is 0.09 lower than the combined reward on training lengths, $p = 0.102$).

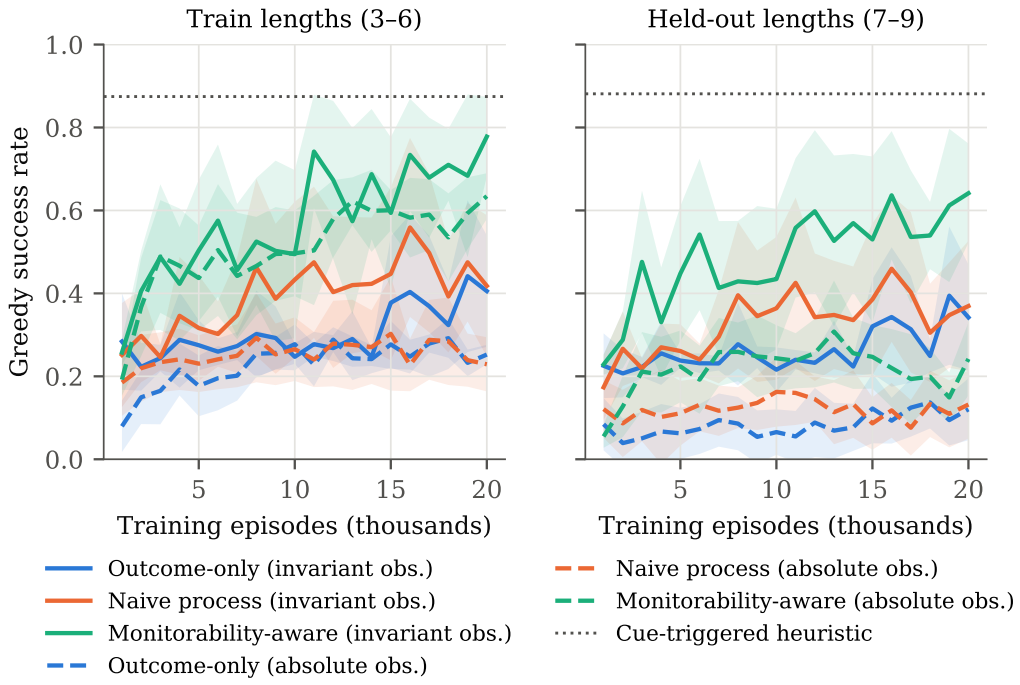


Figure 2: Greedy success during training (mean over 10 seeds, shaded 95% t-interval), evaluated every 1,000 episodes on validation episodes. The mean hides large per-seed swings (Figure 4).

6.2 A shared failure mode: detecting without repairing

A checkpoint can fail even though it still checks. It catches the fault with an UNSAFE result and then does not complete the repair. Late checkpoints that collapse in this way, with success ≤ 0.35 but at least half of all faults caught, occur under every reward: 15% of late checkpoints for outcome-only, 29% for naive process and 21% for monitorability-aware rewards, all with invariant observations. The effect is also visible in evaluated agents (Appendix Table 12). Even on training lengths, where no input is out of range, the last checkpoints of the monitorability-aware agents complete the repair after a caught fault in only 66% (absolute features) and 83% (invariant features) of episodes. The cue-triggered and budget-aware heuristics always complete it (100%).

This is a learning failure, not reward hacking in the sense of a policy that scores well on the specified reward but not on the intended one [Krakovna et al., 2020, Skalse et al., 2022]. Under every reward, completing the repair, whenever the budget still allows it, is worth far more than any detection bonus: success pays $+1$ and failure -1.2, against at most 0.43 for catching a cued fault. A detect-without-repair policy therefore earns less return than a repairing one even under the monitorability-aware reward.

The learned values barely separate repairing from continuing. We probe each checkpoint at the decision points right after an UNSAFE result where the budget still allows the repair, on training-length evaluation episodes (`scripts/q_probe.py`; Appendix Table 17). Points where it does not (both actions then lose) are excluded. Pooled over seeds they are at most 12% of post-UNSAFE points for any probed agent, but they exceed half at 3 of the 295 individual checkpoints (all of them naive agents' last checkpoints, one with no affordable point at all). At each such point we compare the learned values of `REVISE` and `THINK` with their true discounted returns under the same greedy policy, obtained exactly by playing each action out from a copy of the environment. For the selected monitorability-aware checkpoints, the two actions are valued at 0.08 (`REVISE`) and 0.06 (`THINK`), while their true

returns are 0.66 and −1.03: the network values a repair and a doomed continuation almost equally, with errors (learned minus true) of +1.09 for continuing and −0.58 for repairing. The same holds for outcome-only and naive agents and at the last checkpoints, where the error is +0.83 to +1.24 for continuing and −1.18 to −0.33 for repairing (Appendix Table 17). Part of the error is a general offset. At the first decision point of each episode, where the hidden state follows the task prior, the selected checkpoints value the greedy action at 0.20 against a realized return of 0.65; relative to that, the excess error of repairing is −0.13 and that of continuing +1.54. This baseline is rough: the start value bootstraps through later, possibly aliased states, and the offset is close to zero at every last checkpoint (−0.02 to 0.09) and for the memory-bit and 8-action-history agents but not for the pending-bit agents (Appendix Table 17), so it varies with what we are probing. The robust observation is the near-tie. That continuing is the more overvalued action is less informative than it looks, because continuing after an UNSAFE result is rarely the greedy choice, so its value is learned mostly from exploratory actions and extrapolated. With the two values nearly level, the greedy choice is fragile. The selected checkpoints of the three rewards continue after an UNSAFE result at 5%, 2% and 0% of such points, and their last checkpoints at 22%, 28% and 25%. The size of the overvaluation does not by itself predict the failure: it persists, and is sometimes larger, in follow-up variants that do not fail (Appendix Table 17). What separates failing from stable agents is whether the greedy choice after an UNSAFE result flips during training.

The observation hides whether a correction is pending. A likely reason is state aliasing [Singh et al., 1994]. The invariant observation does not record whether a caught fault is still unrepaired, so after continuing past it the agent can see exactly the observations it sees when its running value is correct (for the monitorability-aware agents, 96% of the observations on the continuing branch also occur on track in the same agent's rollouts). Bootstrapped values of continuing are then pulled toward those of on-track states, and the failure that follows shows up only as a slightly lower average outcome of the shared observations, which makes the choice easy to flip. §6.9 tests this directly with a bit that is set by an UNSAFE check and cleared once the located step has been correctly redone.

6.3 Length generalization is mainly a representation problem

Figure 1 separates the effect of the reward from the effect of the observation encoding. With the original absolute encoder, success falls sharply on longer chains for every reward. On lengths 10–14 all three absolute-encoder agents are *worse than never checking*: 0.05, 0.03 and 0.03, versus 0.20. On lengths 10–14 they mostly fail in one specific way: they commit before executing a single step, in 56% of monitorability-aware episodes (only 7% at 7–9, where their failures are more varied). The depth feature $d/9$ exceeds 1 on these chains, and the policy has never seen such an input.

More generally, every observation the absolute encoder produces on held-out lengths is new. We count an observation as unseen if its feature vector does not occur exactly in the reference sample, so this measures novelty rather than distance. Across rollouts of the hand-coded policies, 100% of decision points at lengths 7–9 have an observation vector that does not occur in a large reference sample from lengths 3–6. For the invariant encoder the figure is 2% at 7–9 and 5% at 10–14.

Changing only the observation encoder removes most of the failure. The monitorability-aware agent with invariant features keeps 0.86 [0.83, 0.89] at lengths 7–9 and 0.74 [0.68, 0.80] at 10–14. These are paired improvements over the absolute encoder of +0.55 [0.45, 0.66] and +0.71 [0.62, 0.80], in 10/10 seeds (Holm-adjusted $p = 0.012$). At lengths 7–9 its difference from the cue-triggered heuristic is −0.02 [−0.05, 0.01]. This is not significant ($p = 0.166$), but it does not demonstrate equivalence either. Both remain below the exact planner (0.94).

It is not only the position features. The absolute encoder differs from the invariant one in more than position. Its budget fraction and slack normalisation depend on chain length, and its gap feature, although its meaning is fixed, takes values on long chains that never occur in training. A control encoder keeps all of these but drops the two position features $t/9$ and $d/9$ (Table 4). It no longer commits at the start, but it generalizes almost as badly as the

Table 4: Encoder control (monitorability-aware reward, 10 seeds, 95% t-interval). "Unseen obs." is the fraction of decision points at lengths 10–14 whose observation does not occur in a reference sample from lengths 3–6, both collected from rollouts of the hand-coded policies (a property of the encoder, not of each agent's own visitation); "commit at t=0" is the fraction of episodes at 10–14 that commit before executing a step.

Encoder	Success 3–6	Success 7–9	Success 10–14	Unseen obs. 10–14	Commit at t=0
Monitorability (abs)	0.76 [0.72, 0.80]	0.31 [0.20, 0.41]	0.03 [0.00, 0.07]	100%	0.56
Monitorability (abs., no position)	0.65 [0.56, 0.74]	0.35 [0.26, 0.43]	0.14 [0.09, 0.19]	98%	0.01
Monitorability (inv)	0.91 [0.89, 0.92]	0.86 [0.83, 0.89]	0.74 [0.68, 0.80]	5%	0.00

full absolute encoder: 0.35 [0.26, 0.43] at 7–9 and 0.14 [0.09, 0.19] at 10–14. Its observations are also mostly unseen on held-out lengths (98% at 10–14). The position features are therefore not the main problem; what the control keeps either changes meaning with length or takes values never seen in training, and this control does not separate the two. The two findings of this section and of §6.2 also interact: much of what the length-dependent encoders lose at 7–9 is faults they catch and then fail to repair (29% of episodes for the absolute encoder and 45% for the no-position control, against 2% for the invariant one).

The reward still matters under the invariant encoder. The monitorability-aware reward beats outcome-only at every length: +0.25 at 7–9 (Holm-adjusted $p = 0.012$) and +0.22 at 10–14 ($p = 0.043$). Naive process reward falls between the two in point estimate. With 10 seeds its separation from outcome-only is not resolved on held-out lengths ($p = 0.098$ at 7–9). Its separation from the monitorability-aware reward is resolved at 7–9 ($p = 0.020$) but not at 10–14 (+0.09 [−0.05, 0.22]).

6.4 What the agents learn: checking cued steps

Figure 3 shows how often each agent checks at a decision point whose current step is unchecked, split by whether that step's cue fired. With invariant features all three rewards produce cue-conditional checking. What distinguishes the monitorability-aware reward is *coverage* of cued steps, not avoidance of un-cued ones. It checks at 0.96 of cued decision points versus 0.71 for outcome-only, while checking un-cued ones at a similar rate (0.17 vs. 0.12). Outcome-only agents with absolute features barely condition on the cue at all (0.29 vs. 0.20). With absolute features, the monitorability-aware agent's cue coverage decays on held-out lengths (0.78 at 7–9). With invariant features it is largely preserved (0.89).

6.5 The residual gap: spending slack that long chains need

The invariant agents still lose ground on the longest chains (0.91 → 0.86 → 0.74). Some decline is intrinsic: with the same slack, longer chains have more steps to cover, and even the exact planner falls from 0.99 to 0.94 and 0.90. The cue-triggered heuristic barely declines (0.87 → 0.88 → 0.85), because it never spends slack on un-cued steps. The budget-aware heuristic, tuned on training lengths, degrades like the learned agent (0.97 → 0.82 → 0.68), which suggests that part of the problem is how the budget is allocated rather than the representation. The six units of slack beyond the chain length must cover every cued check plus one repair. Two ways of spending slack are candidates:

- **Un-cued checks.** On short chains, slack is left over for checking un-cued steps, which pays off. On long chains there are more cued steps to cover, and slack spent early is missing later. The budget-aware heuristic loses ground in exactly this way.
- **Non-repair `REVISE`s.** The learned agents also spend slack on `REVISE`s that do not follow an UNSAFE check (0.74 per episode at 10–14, each costing two units with its re-execution). This is a consequence of the memoryless observation (§3): after a SAFE check, undoing and redoing a step leads to the same observation as a genuine repair.

Fault detection at 10–14 is 0.75 for the learned agent and 0.60 for the budget-aware heuristic, against 0.86 for the cue-triggered one. The exact planner shows that the best allocation

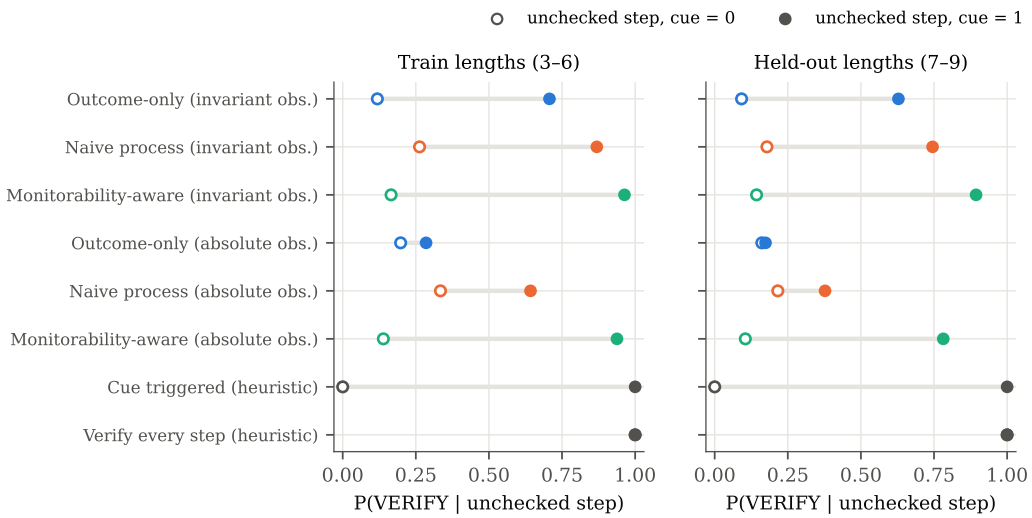


Figure 3: Cue-conditional checking: probability of choosing `VERIFY` at a decision point whose current step is unchecked, when that step's cue is off (open) or on (filled). The cue-triggered heuristic is perfectly cue-conditional by construction.

depends on how many steps remain. At the same slack and cue, whether checking a step is optimal changes with the number of remaining steps (Appendix Figure 5): with most of the slack left, the planner checks un-cued steps when few steps remain but not when many do. A policy that sees only slack and cue, as the invariant encoder provides, cannot follow this rule at every length. Which of the two ways of spending slack costs the learned agents most is still a hypothesis: §6.9 removes most non-repair `REVISE`s with a memory bit and finds no held-out gain, so that part of the explanation is at best partial.

6.6 Robustness to environment shifts

Under shifts at lengths 7–9 that the agents were not trained on (Appendix E), the invariant monitorability-aware agent is robust to the fault rate and tracks the cue-triggered heuristic when cue quality changes, but it degrades when checks cost more than in training and when the budget slack changes in either direction.

6.7 Which reward terms matter

Table 5 removes one group of terms at a time from the monitorability-aware reward, with invariant observations and 10 seeds each. Every variant is evaluated on the same episodes as the full reward, so the differences below are paired.

Every variant that pays for checks learns to check; none needs a particular term. Every variant that keeps the per-action cost and *some* payment for informative checks reaches 0.88–0.89 success on training lengths at its selected checkpoint. That holds with the cue terms removed, with the safe/fault check bonuses removed, with the cue bonus alone, and with the cue, gap and waste terms all removed (Table 5), against 0.72 for outcome-only. None of the variants changes how the agents fail during training. Apart from the variant without `REVISE`, which cannot repair at all, collapsed checkpoints almost never fall into the low-detection regime (at most 1% of late checkpoints). Every one of these variants shows the detect-without-repair collapse of §6.2, in 12% to 39% of late checkpoints. The collapse therefore does not hinge on any one term.

Generalization depends on not over-checking. On longer chains the variants separate.

Table 5: Reward ablations (invariant observations; mean over 10 seeds, 95% t-interval, selected checkpoints; Appendix Table 11 repeats this for last checkpoints). $P(V \mid c)$: probability of checking at a decision point whose current step is unchecked with cue c, at lengths 7–9.

Variant	Success 3–6	Success 7–9	Success 10–14	$P(V \mid c{=}1)$	$P(V \mid c{=}0)$
Full reward	0.91 [0.89, 0.92]	0.86 [0.83, 0.89]	0.74 [0.68, 0.80]	0.89	0.14
− cue bonus/penalty	0.88 [0.85, 0.90]	0.64 [0.56, 0.72]	0.40 [0.30, 0.51]	0.83	0.30
− unchecked-gap term	0.89 [0.87, 0.92]	0.84 [0.81, 0.87]	0.73 [0.61, 0.86]	0.88	0.13
− wasted-check penalty	0.89 [0.87, 0.92]	0.85 [0.81, 0.89]	0.68 [0.53, 0.82]	0.92	0.15
− safe/fault check bonuses	0.89 [0.88, 0.90]	0.78 [0.75, 0.82]	0.65 [0.56, 0.74]	0.92	0.15
− cue, gap and waste terms	0.89 [0.87, 0.90]	0.79 [0.72, 0.85]	0.66 [0.55, 0.76]	0.83	0.16
cue bonus only	0.88 [0.86, 0.90]	0.85 [0.82, 0.88]	0.77 [0.72, 0.83]	0.92	0.10
− per-action cost	0.85 [0.80, 0.89]	0.76 [0.69, 0.83]	0.61 [0.50, 0.71]	0.92	0.11
REVISE disabled	0.20 [0.19, 0.21]	0.19 [0.16, 0.21]	0.15 [0.12, 0.19]	0.61	0.11

- **Cue bonus alone.** With every other process term removed, the cue bonus is indistinguishable from the full reward (0.85 at 7–9 and 0.77 at 10–14, vs. 0.86 and 0.74).
- **Removing only the cue terms.** At selected checkpoints this is the most damaging change: the full reward beats this variant by 0.22 [0.13, 0.32] at 7–9 and by 0.33 at 10–14 ($p = 0.004$). At 10–14 its mean (0.40) is below that of outcome-only agents (0.52), but that difference is not resolved (-0.11 [-0.26, 0.03], $p = 0.121$). Removing the cue terms *together with* the gap and waste terms costs less: the full reward's advantage is then 0.07 at 7–9, and the variant still beats outcome-only there by 0.17 ($p = 0.008$).
- **The interaction.** The gap term refunds any check that closes the unchecked gap, cued or not. Without the cue terms to counterbalance it, agents check un-cued steps twice as often (0.30 vs. 0.14 at 7–9). That costs little on short chains but spends the slack that long chains need (§6.5).
- **Other terms.** The full reward also beats the variant without the per-action cost on held-out lengths (by 0.10, $p = 0.002$ at 7–9), and the variant without the safe/fault check bonuses (by 0.08). Removing the gap term or the waste penalty alone has no resolvable effect at selected checkpoints.
- **No `REVISE`.** Without `REVISE`, a caught fault cannot be repaired and success falls to the never-check level (0.20).

The ranking does not survive last-checkpoint evaluation. At the last checkpoint (Appendix Table 11) training-length success spreads from 0.55 to 0.78 across the same variants, and removing the cue terms is no longer uniquely damaging. At 7–9 the full reward's last checkpoints beat those without the cue terms by 0.19 ($p = 0.027$), and, in point estimate, those without the per-action cost by 0.20 ($p = 0.061$) and without the waste penalty by 0.15 ($p = 0.127$). The ablation results above therefore describe the best checkpoints each reward produces, not where training ends.

6.8 Training stability and sensitivity

DQN agents in this environment are unstable across checkpoints. This is related to the policy churn of Schaul et al. [2022], which they measure as changes in greedy actions between consecutive updates; we measure it as swings in evaluated performance over thousands of episodes. Figure 4 plots greedy success for each seed of the two monitorability-aware conditions. Policies repeatedly switch, within a few thousand episodes, between a solution that checks and repairs and one that checks but does not repair (§6.2). The probe of §6.2 suggests why: the two solutions differ in one greedy choice whose learned values are nearly tied. After the first 5,000 episodes, the mean absolute change in training-length success between consecutive evaluations is 0.23 for the invariant monitorability-aware agent. That is larger than for outcome-only agents (0.10), which mostly sit in the low-detection regime.

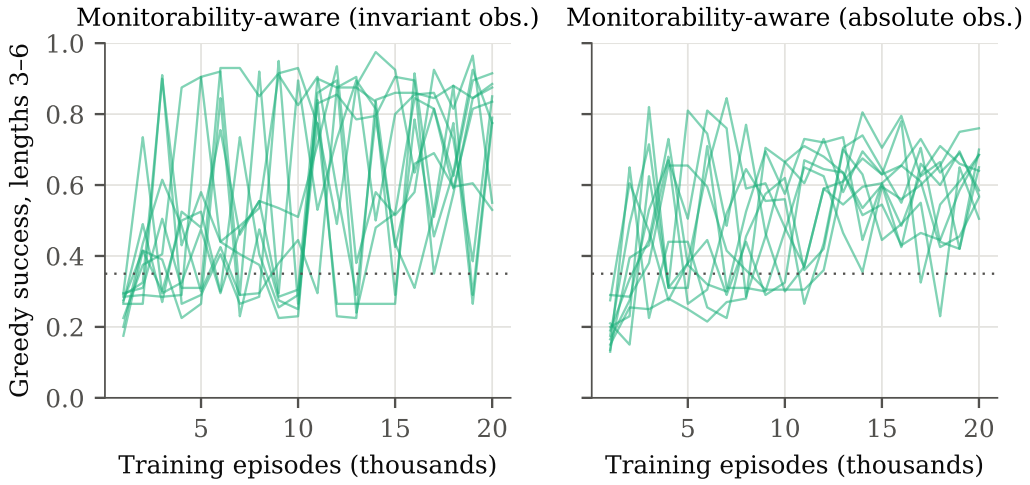


Figure 4: Greedy success on training lengths for each of the 10 seeds of the monitorability-aware agents. The dotted line marks success 0.35. Individual runs jump between solutions throughout training; for these agents the low regime is detect-without-repair, not low detection (§6.2).

Because of this churn, last-checkpoint results (Appendix Table 10) are lower and noisier than selected-checkpoint results. For example, the invariant monitorability-aware agent scores 0.63 [0.51, 0.76] instead of 0.86 [0.83, 0.89] at lengths 7–9. With last checkpoints, the results split three ways:

- **Unchanged:** the encoder gap (+0.49 at 10–14, $p = 0.002$) and the advantage of the monitorability-aware reward over outcome-only (+0.30 at 7–9, $p = 0.004$).
- **Weaker:** the difference between naive process reward and outcome-only, which was not clearly resolved at selected checkpoints either ($p = 0.098$ at 7–9), shrinks to +0.03 [−0.16, 0.22].
- **Lost:** closeness to the cue-triggered heuristic. The last checkpoint's difference from it is −0.25 [−0.37, −0.12] at lengths 7–9, and the ranking of the reward ablations changes (§6.7).

Selection uses only training-length validation episodes, so it cannot favour the encoder that generalizes better.

Sensitivity to the learner's hyperparameters. Under three alternative DQN settings (learning rate $3{\cdot}10^{-4}$, hard target updates every 1,000 steps, Polyak-averaged targets) and with twice the training, five seeds each (Appendix F), the encoder gap reproduces under each of the three alternative settings (the longer runs use only the invariant encoder), and in all four settings outcome-only agents' late checkpoints collapse in 81%–96% of cases against 17%–22% for the monitorability-aware agents. None of the settings removes the churn.

6.9 Follow-up interventions

We test the explanations suggested by the analyses above. Each intervention is added to the monitorability-aware agent with invariant observations, 10 seeds each, and compared with the matching reference agent on paired episodes (Tables 6 and 7); the repair-pending bit and the action histories are also trained under the other two rewards. These comparisons are exploratory and their p-values are not corrected for multiplicity. In brief: a bit that records whether a caught fault is still unrepaired removes the detect-without-repair collapse under every reward. A history of the agent's own last 3 actions all but removes collapse under every reward and, under the monitorability-aware reward, generalizes to long chains at the last checkpoint, whereas a history of the last 8 actions, which on training lengths also reveals

Table 6: Follow-up interventions (mean over 10 seeds, 95% t-interval). "Last ckpt." evaluates the last checkpoint instead of the selected one; "collapsed" is the fraction of late checkpoints (after 5,000 episodes) with training-length success ≤ 0.35; "wasted `REVISE`" counts non-repair `REVISE`s per episode at lengths 10–14.

Variant	Success 3–6	Success 7–9	Success 10–14	Last ckpt. 7–9	Collapsed	Wasted REVISE
Monitorability (inv)	0.91 [0.89, 0.92]	0.86 [0.83, 0.89]	0.74 [0.68, 0.80]	0.63 [0.51, 0.76]	21%	0.74
+ slack-per-step feature	0.83 [0.80, 0.87]	0.46 [0.35, 0.57]	0.28 [0.20, 0.35]	0.37 [0.31, 0.43]	38%	0.64
+ fault-located memory bit	0.94 [0.92, 0.95]	0.88 [0.86, 0.91]	0.74 [0.65, 0.83]	0.81 [0.70, 0.91]	10%	0.23
fault bonus paid on repair	0.89 [0.88, 0.91]	0.79 [0.71, 0.87]	0.64 [0.52, 0.75]	0.75 [0.67, 0.83]	1%	0.51
memory bit + bonus on repair	0.93 [0.92, 0.95]	0.83 [0.76, 0.91]	0.66 [0.54, 0.77]	0.79 [0.72, 0.85]	0%	0.15
+ repair-pending bit	0.95 [0.94, 0.95]	0.83 [0.76, 0.91]	0.60 [0.45, 0.75]	0.89 [0.87, 0.91]	0%	0.08
+ last 8 actions	0.97 [0.96, 0.98]	0.78 [0.74, 0.81]	0.41 [0.30, 0.51]	0.75 [0.69, 0.81]	0%	0.31
+ last 3 actions	0.96 [0.96, 0.97]	0.85 [0.80, 0.90]	0.66 [0.58, 0.75]	0.91 [0.90, 0.92]	0%	0.01

position, fails on long chains, where many of its windows are unfamiliar. Paying the fault bonus at the repair step is a reward-side remedy for the same failure. A memory of having seen a fault removes wasted `REVISE`s but not the failure, and a slack-per-remaining-step feature made the repair decision worse (Appendix G).

A repair-pending bit removes the failure under every reward. This bit is set by an UNSAFE check and cleared once the located step has been correctly re-executed, so it records exactly what the invariant observation hides; an agent that remembered its own checks and undos could compute it. We train it with all three rewards (Table 7). Late checkpoints that detect faults without repairing them fall from 15%, 29% and 21% (outcome-only, naive and monitorability-aware) to 0%, 0% and 0%. The greedy choice after an UNSAFE result becomes stable: the last checkpoints continue past a caught fault at 0%, 1% and 0% of such points. The bit does not make the values accurate (continuing is still overvalued by 1.07 for the monitorability-aware agents). Removing the aliasing is therefore enough to stabilise the choice, but aliasing is not sufficient for instability: the repair payment below also stabilises it, although with that reward 88% of the continuing-branch observations still occur on track.

The bit also changes how often outcome-only agents collapse at all. Outcome-only agents now collapse in 20% [13%, 27%] of late checkpoints instead of 85%, and fall into the low-detection regime in 15% instead of 60%, presumably because checks pay off only once the repair that follows them is learned reliably. What this means for check payments depends on the metric. Their advantage in collapse shrinks with the bit (20% against 0% of late checkpoints, from 85% against 21% without it), so part of what they buy in §6.1 is compensation for the missing information. Their gain in late training-length success does not shrink: with the bit, outcome-only agents behave much like monitorability-aware agents without it (collapse 20% against 21%; mean late training-length success 0.63 against 0.63), and the check payment still adds about as much on top (0.93). With the bit, the check payment shows up in training stability rather than in peak performance. At selected checkpoints the two rewards are indistinguishable on held-out lengths (monitorability-aware minus outcome-only: 0.04 [−0.02, 0.10] at 7–9 and −0.06 at 10–14). But outcome-only agents still collapse more often (20% of late checkpoints against 0%), and their last checkpoints are much worse (0.37 against 0.89 at 7–9, $p = 0.002$), already on training lengths (0.62 against 0.92), so only part of that gap is a generalization deficit. With the bit, the monitorability-aware agents' last checkpoints are slightly worse than their selected ones on training lengths (−0.03, 8/10 seeds lower), better on held-out lengths (+0.22 at 10–14, $p = 0.004$), and better than the reference's last checkpoints (by 0.26 at 7–9, $p = 0.002$). At selected checkpoints, by contrast, the bit lowers the monitorability-aware agents' success on the longest chains, most likely because selection on saturated training-length validation picks checkpoints that over-check (Appendix G).

A short history of recent actions all but removes collapse; a long one fails on long chains. The pending bit is computed by the environment. To test whether a learner can do without it, we instead add the agent's recent actions and check results to the invariant observation, one-hot coded and zero-padded at the start of an episode: either the last 8 actions or only the last 3 (Table 7; the 3-action window was added after seeing the 8-action results). The two windows carry different information. Three actions show a caught fault until it is repaired or for two steps after it is skipped, and whether a `REVISE` followed a

SAFE or an UNSAFE check, but little about position. Eight actions hold the whole episode so far at 86% of training-length decision points, so they also reveal position (the number of steps executed) and the full record of checks, which the invariant encoder was built to hide. Neither window shows the chain length, so neither says much more than the invariant observation about how many steps remain. The networks also differ in size (5,060 parameters for the invariant observation, 6,020 with 3 actions and 7,620 with 8).

On training lengths both windows all but remove collapse under every reward (0%, 2% and 0% of late checkpoints with 3 actions, for outcome-only, naive and monitorability-aware rewards; 0%, 0% and 0% with 8), more completely than the pending bit does for outcome-only agents (20%). Selected checkpoints reach 0.95–0.97 training-length success. The check payment's advantage in late training-length success shrinks as the observation carries more: it is +0.31 without any addition, +0.29 with the pending bit, +0.13 [0.09, 0.16] with 3 actions and +0.04 with 8, positive in 10/10 seeds even then ($p = 0.002$); with 8 actions the check-paid agents also still first exceed 0.5 earlier (median episode 1,000 against 2,500, $p = 0.004$). Most of the shrinkage thus already happens with the short window, so position and the full record of checks can account for at most the remaining difference between the two windows. Absolute gaps are also compressed near the ceiling. As a ratio of late failure rates, outcome-only agents fail 1.83 times as often as check-paid ones without any addition and 1.79 times as often with 8 actions (4.95 with the pending bit, 3.01 with 3 actions), so on that scale no observation removes the payment's advantage. The value probe shows that the longer window changes what the agents learn: after a caught fault, the 3-action agents overvalue continuing about as much as the pending-bit agents (excess errors 1.47, 1.35 and 1.37), the 8-action agents much less (0.16, 0.24 and 0.40).

On long chains the short window holds up better at the last checkpoint, but only the monitorability-aware agents actually transfer better. At 10–14 the 3-action agents' last checkpoints beat the same rewards without a history by +0.20, +0.22 and +0.30. For outcome-only and naive rewards this mostly reflects training stability: the lead is larger on training lengths (+0.47 and +0.49), where the agents without a history have mostly collapsed, and the 3-action agents fall from 0.86 and 0.90 to 0.50 and 0.54. The monitorability-aware ones fall only from 0.93 to 0.82 [0.78, 0.86], about as well as with the pending bit (0.83) and within an unresolved −0.03 [−0.07, 0.00] of the cue-triggered heuristic (a comparison made after seeing the results). At selected checkpoints no 3-action agent does better than the agents without a history at 10–14 (0.04, −0.15 and −0.07, none resolved); for the monitorability-aware agents this is most likely the selection effect described for the pending bit (selected 0.66, last 0.82). With 8 actions, even fault-free chains succeed only 0.35, 0.43 and 0.60 of the time at 10–14, and 68%, 52% and 40% of episodes end before the chain is complete (with 3 actions, 5%, 7% and 5%). Most of these episodes run out of budget; the outcome-only agents with 8 actions also commit early with budget left in 23% of episodes.

The agents' own rollouts show how much more often the long window meets observations it has not seen (`scripts/agent_coverage.py`; an observation counts as unseen if it never occurs in the same checkpoint's greedy training-length rollouts). At 10–14, 39% of the monitorability-aware 8-action agents' decision points are unseen at the selected checkpoints and 41% at the last ones, against 14% and 10% with 3 actions and 2% and 3% for the invariant reference. At points with no slack left, 83% of the 8-action agents' observations are unseen (3 actions: 53%; invariant: 13%; selected checkpoints). Both windows over-check early in long chains (at the selected checkpoints the monitorability-aware agents check 53% of un-cued steps in the first half of the chain with 8 actions and 49% with 3, against 13%). Once no slack is left, any check or undo makes finishing impossible; the 8-action agents still check or undo at 9% of such points (4% at the last checkpoint), the 3-action agents at 0%. Some of that spending comes in episodes that are already lost, for instance after a fault caught too late to repair, so it is not by itself the cause. For a greedy policy, spending at familiar observations is fixed by its training-length behaviour, so comparisons of familiar with unseen points within one agent say little; what differs between the windows is how often the critical points are unseen. The long window's failure is consistent with the coverage account of §6.3, but this analysis does not show how the novelty turns into failure. A short memory of its own recent actions gives the learner what the pending bit gives, and for outcome-only agents more, without the hand-made state.

Table 7: Each reward with the invariant observation, with the repair-pending bit added, and with the last 8 or the last 3 actions added instead (10 seeds, 95% t-intervals). Collapse columns, each a fraction of *all* late checkpoints (after 5,000 episodes): collapsed (training-length success ≤ 0.35); collapsed and detecting without repairing (fault detection ≥ 0.5); collapsed with low detection (below 0.2: the checkpoint checks, but almost never catches a fault). Collapsed checkpoints with intermediate detection are in neither of the last two. "Late success": mean training-length success of the late checkpoints. Bottom: success at selected and last checkpoints.

Reward and observation	Collapsed	no repair	low detection	Late success
Outcome-only	85%	15%	60%	0.32
Outcome-only + pending bit	20%	0%	15%	0.63
Outcome-only + last 8 actions	0%	0%	0%	0.92
Outcome-only + last 3 actions	0%	0%	0%	0.81
Naive process	62%	29%	17%	0.43
Naive process + pending bit	15%	0%	7%	0.69
Naive process + last 8 actions	0%	0%	0%	0.93
Naive process + last 3 actions	2%	0%	2%	0.85
Monitorability-aware	21%	21%	0%	0.63
Monitorability-aware + pending bit	0%	0%	0%	0.93
Monitorability-aware + last 8 actions	0%	0%	0%	0.95
Monitorability-aware + last 3 actions	0%	0%	0%	0.94

Reward and observation	Success 3–6	Success 7–9	Success 10–14	Last ckpt. 7–9	Last ckpt. 10–14
Outcome-only	0.72 [0.64, 0.81]	0.62 [0.48, 0.75]	0.52 [0.36, 0.67]	0.33 [0.21, 0.46]	0.30 [0.19, 0.41]
Outcome-only + pending bit	0.90 [0.89, 0.92]	0.79 [0.71, 0.87]	0.66 [0.53, 0.78]	0.37 [0.28, 0.46]	0.27 [0.22, 0.33]
Outcome-only + last 8 actions	0.97 [0.97, 0.97]	0.69 [0.65, 0.74]	0.22 [0.15, 0.29]	0.69 [0.60, 0.79]	0.29 [0.19, 0.39]
Outcome-only + last 3 actions	0.95 [0.93, 0.97]	0.80 [0.75, 0.85]	0.56 [0.45, 0.68]	0.68 [0.56, 0.80]	0.50 [0.41, 0.59]
Naive process	0.85 [0.80, 0.91]	0.76 [0.64, 0.87]	0.65 [0.52, 0.78]	0.36 [0.21, 0.51]	0.31 [0.20, 0.43]
Naive process + pending bit	0.92 [0.91, 0.93]	0.75 [0.65, 0.84]	0.55 [0.44, 0.66]	0.39 [0.33, 0.46]	0.30 [0.26, 0.34]
Naive process + last 8 actions	0.97 [0.96, 0.97]	0.70 [0.67, 0.73]	0.29 [0.19, 0.38]	0.74 [0.70, 0.78]	0.36 [0.27, 0.44]
Naive process + last 3 actions	0.97 [0.96, 0.98]	0.78 [0.73, 0.82]	0.50 [0.38, 0.61]	0.73 [0.61, 0.85]	0.54 [0.36, 0.71]
Monitorability-aware	0.91 [0.89, 0.92]	0.86 [0.83, 0.89]	0.74 [0.68, 0.80]	0.63 [0.51, 0.76]	0.52 [0.41, 0.64]
Monitorability-aware + pending bit	0.95 [0.94, 0.95]	0.83 [0.76, 0.91]	0.60 [0.45, 0.75]	0.89 [0.87, 0.91]	0.83 [0.76, 0.89]
Monitorability-aware + last 8 actions	0.97 [0.96, 0.98]	0.78 [0.74, 0.81]	0.41 [0.30, 0.51]	0.75 [0.69, 0.81]	0.45 [0.38, 0.52]
Monitorability-aware + last 3 actions	0.96 [0.96, 0.97]	0.85 [0.80, 0.90]	0.66 [0.58, 0.75]	0.91 [0.90, 0.92]	0.82 [0.78, 0.86]

Paying at the repair step removes the detect-without-repair collapse. Moving the 0.25 fault bonus from detection to the first correct re-execution of the located fault places a reward at the repair step itself. If the collapse of §6.2 is a credit-assignment failure, this should help. The test changes two things at once: it removes the 0.25 detection payment and adds a repair payment, and we do not separate them. The cue bonus is still paid when a cued fault is caught. The ablation that removes the detection payment (together with the safe-check bonus) *without* adding a repair payment (no check bonuses, §6.7) still collapses in 13% [8%, 17%] of late checkpoints. The repair payment, not the missing detection payment, is therefore the likely active ingredient. Among late checkpoints (after 5,000 episodes), collapsed ones drop from 21% [10%, 32%] to 1% [0%, 2%]. The mean absolute change between consecutive evaluations drops from 0.23 to 0.06. With the memory bit added as well, collapse disappears entirely (0%). This is a reward-side remedy for the same failure; we did not try it with the other rewards. Its selected checkpoints come early (median episode 4,500, and 6/10 within the first 5,000 episodes, which the collapse statistic excludes), and on long chains some of them still fail to repair: success after a detection at 10–14 is 0.75, against 0.91 for the reference. Its benefit shows up in reliability rather than peak performance: at selected checkpoints it does not beat the original reward, at the last checkpoint it does (by +0.12 at 7–9, $p = 0.016$; Appendix G).

7 Discussion

What rewarding checks buys. Paying for informative checks did not unlock behaviour that outcome reward can never reach: with a suitable observation, outcome-only agents also find checking policies. What it bought was speed and retention. Outcome-only agents

found checking policies late, mostly lost them again, and did not catch up with twice the training. Agents under the naive process reward collapsed less often but still in most late checkpoints. Its step-correctness bonus alone, the term closest to a process reward model, bought more stability the larger it was, and not through the undo-and-redo loop that a repeatable step bonus invites. At its largest, and still farmable, it prevented collapse about as well as a substantial check payment; paid once per step it collapsed more often, and either way its checkpoints plateaued lower, with no resolved gain at selected checkpoints. In the main and sensitivity experiments, agents paid substantially for checks never fell into checking that misses faults (some follow-up and ablation variants did, in at most 5% of late checkpoints, apart from the ablation without `REVISE`, which cannot repair). Part of this benefit, though, compensates for information the invariant observation hides (see below). The contrast between outcome-only and substantial payment held under every learner setting we tried; the held-out benefit of the naive process reward did not. On training lengths, most of that benefit came from paying for checks at all, not from any specific term (§6.7). What decided whether the learned checking worked on long chains was its *allocation*. The gap term, meant to keep traces diagnosable, rewards any check that closes an unchecked stretch. At selected checkpoints, agents without the cue terms to counterbalance it over-checked and ran out of budget on long chains; this interaction does not reproduce at the last checkpoint, and we did not isolate the gap term's part in it. Step-level rewards shape not only whether an agent checks but how it spends a shared budget, and in this testbed that mattered more for generalization than whether checking was learned at all.

What the observation records matters as much as what the reward pays. Agents under every reward drifted into policies that catch faults and then fail to repair them. This is a learning failure, not specification gaming, because repairing is worth far more than any detection bonus. A single bit recording that a correction is still pending removed the failure under every reward and made the greedy choice stable, although the learned values stayed miscalibrated. Under the monitorability-aware reward, a reward-side change (paying the fault bonus at the repair step rather than at detection) also stabilised the choice, and an observation-side change (a memory of having seen a fault) helped less clearly. For learners with limited state, the lesson is that a reward for noticing a problem does little if the state does not record that the problem is still open; either the state should carry it or the reward should pay for completing the correction. Language-model reasoners condition on their whole trace, so for them the question is whether such state is learned from history. Agents that saw only their last 3 actions learned it: at most 2% of their late checkpoints collapsed under any reward, and under the monitorability-aware reward their last checkpoints kept most of their performance on longer chains. Agents that saw their last 8 actions also stopped collapsing but generalized poorly. On training lengths that window holds almost the whole episode; on long chains many of its windows, and most of those at low-slack points, are ones the policy never meets in its own training-length rollouts (§6.9). We have not identified exactly how that makes it fail. In our small MLP agents, then, the longer window reintroduced the novelty that the invariant encoder removed and the 3-action window largely avoided; whether the same holds for learned summaries of a full history, as in a language model's trace, is open. The bit changed the case for check payments only in part. It narrowed their advantage in collapse, but with the bit, outcome-only late checkpoints still trailed the check-paying agents' by about as much as without it (0.63 against 0.93 mean training-length success). Recent actions narrowed that gap further, most of the way already with 3 actions (0.81 against 0.94) and almost completely with 8 (0.92 against 0.95). Near the ceiling, though, absolute gaps are compressed. As a ratio, outcome-only late checkpoints failed 1.83, 4.95, 3.01 and 1.79 times as often as check-paid ones (invariant observation, pending bit, 3 and 8 actions), never less than 1.79. Check payments thus helped under every observation we tried, by less in absolute terms the more the observation carried.

What rewarding checks does not buy. Process rewards did not fix length generalization. With the original encoder's length-dependent features, the best reward still fell below the never-check baseline on long chains, where every observation was novel to the policy. Removing only the two position features was not enough. Two kinds of features plausibly cause trouble: features whose mapping depends on the chain length (the budget fraction and the slack scaled by the total budget), and features with a fixed meaning that take values

on long chains never seen in training (the gap normalised by a fixed constant, and, in the follow-ups, slack per remaining step). The invariant encoder avoids the first kind and caps the second (which also changes the gap feature on training lengths). Our encoder control removes only the position features and keeps both kinds, so we have not separated them. A feature that adds the missing remaining-length information did not help either; it mainly destabilised the repair decision. For process-supervised reasoning models, this suggests that "learned to verify on training problems" and "will verify appropriately on longer problems" are separate claims, to be evaluated separately. Length-dependent features are a plausible way for the second to fail while the first holds.

Evaluation lessons. Two problems in the earlier version of this study pushed its conclusion toward a false negative: a representation confound, and last-checkpoint evaluation of unstable policies. A third, a localization metric that credited unchecked faults, inflated diagnostic scores without changing the main comparison. None of the three was visible in the headline numbers. We recommend four practices for RL on reasoning-like tasks:

- report learning curves and per-seed variability;
- select checkpoints only on in-distribution validation;
- measure how much of a held-out set produces inputs the policy never saw in training;
- test whether out-of-distribution failures move when the representation changes but the reward does not.

Limitations. The environment is deliberately minimal. It has at most one fault, checks are exact, the cue is given rather than learned, and the agent is a small MLP rather than a language model. The "monitorability-aware" reward is a proxy with no monitor behind it. Process rewards use privileged step correctness, standing in for an idealised process reward model. The terms that help on long chains, the cue terms, encode the same knowledge as the cue-triggered heuristic. Their benefit is therefore evidence that well-designed shaping helps, not that a generic step-correctness reward model would: the naive process reward and its step term, which are closer to one, made training more stable but did not resolvably raise held-out success at selected checkpoints, even with a step bonus seven times the original. A step bonus that large also makes redoing correct steps profitable unless it is paid once per step. The exact planner is proven optimal only among policies that never go back to check an earlier step; we verified that this restriction does not bind on chains of up to 7 steps, not beyond. We study one learner (Double DQN) with a small sensitivity sweep; policy-gradient or recurrent agents might be more stable or allocate their budget differently. The encoders differ in feature scaling as well as in length dependence. The coverage statistic counts exact matches, so it measures novelty, not distance: many unseen held-out observations of the no-position control lie close to training observations. It is consistent with the held-out differences but does not by itself explain them, nor why the absolute encoders also learn worse on training lengths. With 10 seeds, differences of a few points are not resolved. Only a family of six headline comparisons, chosen after the results, is corrected for multiplicity; the remaining p-values should be read as exploratory. Finally, at selected checkpoints no encoder we tried closes the gap to the cue-triggered heuristic on the longest chains; only the last checkpoints of the monitorability-aware agents with the repair-pending bit or the 3-action history come within an unresolved difference of it (-0.02 and -0.03; comparisons made after seeing the results).

8 Conclusion

In a small budgeted-verification testbed, rewarding informative checks helped agents find checking policies sooner and keep them. Whether the learned checking carried over to longer chains depended mostly on the observation encoder, and on whether the reward kept checking frugal, more than on whether checking was learned at all. A failure shared by every reward, catching faults without repairing them, disappeared under every reward once the observation recorded whether a caught fault was still open, or showed the agent its last few actions. Correcting a representation confound turned an apparent negative result about

process-rewarded RL into a positive but qualified one. We hope the testbed is useful for probing how process supervision shapes *when* learned reasoners check their work.

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Reproducibility. All code, configs, trained checkpoints, learning curves and per-seed evaluation results are in the accompanying repository (https://github.com/raj200501/UltraRL).

- `make report` and `make paper` rebuild every figure, table and generated number in this paper from the saved results, and CI checks that the committed figures, tables and numbers match byte for byte.
- `make reproduce` retrains every agent from scratch, about 8 hours on a 4-core CPU, and checks the checkpoints bitwise against the saved ones (with the pinned package versions on an AVX-512 CPU; Appendix B gives a statistical check for other hardware).
- `make equivalence` compares the environment with the original implementation, which is vendored in the repository.

References

Rishabh Agarwal, Max Schwarzer, Pablo Samuel Castro, Aaron Courville, and Marc G. Bellemare. Deep reinforcement learning at the edge of the statistical precipice. In *Advances in Neural Information Processing Systems*, 2021.

Cem Anil, Yuhuai Wu, Anders Andreassen, Aitor Lewkowycz, Vedant Misra, Vinay Ramasesh, Ambrose Slone, Guy Gur-Ari, Ethan Dyer, and Behnam Neyshabur. Exploring length generalization in large language models. In *Advances in Neural Information Processing Systems*, 2022.

Bowen Baker, Joost Huizinga, Leo Gao, Zehao Dou, Melody Y. Guan, Aleksander Madry, Wojciech Zaremba, Jakub Pachocki, and David Farhi. Monitoring reasoning models for misbehavior and the risks of promoting obfuscation. *arXiv preprint arXiv:2503.11926*, 2025.

Colin Bellinger, Rory Coles, Mark Crowley, and Isaac Tamblyn. Active measure reinforcement learning for observation cost minimization. In *Canadian Conference on Artificial Intelligence*, 2021.

Frederick Callaway, Sayan Gul, Paul M. Krueger, Thomas L. Griffiths, and Falk Lieder. Learning to select computations. In *Conference on Uncertainty in Artificial Intelligence*, 2018.

Yanda Chen, Joe Benton, Ansh Radhakrishnan, Jonathan Uesato, Carson Denison, John Schulman, Arushi Somani, Peter Hase, Misha Wagner, Fabien Roger, Vlad Mikulik, Samuel R. Bowman, Jan Leike, Jared Kaplan, and Ethan Perez. Reasoning models don't always say what they think. *arXiv preprint arXiv:2505.05410*, 2025.

Karl Cobbe, Oleg Klimov, Chris Hesse, Taehoon Kim, and John Schulman. Quantifying generalization in reinforcement learning. In *International Conference on Machine Learning*, 2019.

DeepSeek-AI. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.

Grégoire Delétang, Anian Ruoss, Jordi Grau-Moya, Tim Genewein, Li Kevin Wenliang, Elliot Catt, Chris Cundy, Marcus Hutter, Shane Legg, Joel Veness, and Pedro A. Ortega. Neural networks and the Chomsky hierarchy. In *International Conference on Learning Representations*, 2023.

Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall, 1993.

Jiaxuan Gao, Shusheng Xu, Wenjie Ye, Weilin Liu, Chuyi He, Wei Fu, Zhiyu Mei, Guangju Wang, and Yi Wu. On designing effective RL reward at training time for LLM reasoning. *arXiv preprint arXiv:2410.15115*, 2024.

Alex Graves. Adaptive computation time for recurrent neural networks. *arXiv preprint arXiv:1603.08983*, 2016.

Matthew Hausknecht and Peter Stone. Deep recurrent Q-learning for partially observable MDPs. In *AAAI Fall Symposium on Sequential Decision Making for Intelligent Agents*, 2015.

Nicholas Hay, Stuart Russell, David Tolpin, and Solomon Eyal Shimony. Selecting computations: Theory and applications. In *Conference on Uncertainty in Artificial Intelligence*, 2012.

Peter Henderson, Riashat Islam, Philip Bachman, Joelle Pineau, Doina Precup, and David Meger. Deep reinforcement learning that matters. In *AAAI Conference on Artificial Intelligence*, 2018.

Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, and Denny Zhou. Large language models cannot self-correct reasoning yet. In *International Conference on Learning Representations*, 2024.

Robert Kirk, Amy Zhang, Edward Grefenstette, and Tim Rocktäschel. A survey of zero-shot generalisation in deep reinforcement learning. *Journal of Artificial Intelligence Research*, 76:201–264, 2023.

Tomek Korbak, Mikita Balesni, et al. Chain of thought monitorability: A new and fragile opportunity for AI safety. *arXiv preprint arXiv:2507.11473*, 2025.

Victoria Krakovna, Jonathan Uesato, Vladimir Mikulik, Matthew Rahtz, Tom Everitt, Ramana Kumar, Zac Kenton, Jan Leike, and Shane Legg. Specification gaming: the flip side of AI ingenuity. DeepMind blog, https://deepmind.google/discover/blog/specification-gaming-the-flip-side-of-ai-ingenuity/, 2020.

Aviral Kumar, Vincent Zhuang, Rishabh Agarwal, Yi Su, John D. Co-Reyes, Avi Singh, Kate Baumli, Shariq Iqbal, Colton Bishop, Rebecca Roelofs, et al. Training language models to self-correct via reinforcement learning. *arXiv preprint arXiv:2409.12917*, 2024.

Tamera Lanham, Anna Chen, Ansh Radhakrishnan, Benoit Steiner, Carson Denison, Danny Hernandez, Dustin Li, Esin Durmus, Evan Hubinger, Jackson Kernion, et al. Measuring faithfulness in chain-of-thought reasoning. *arXiv preprint arXiv:2307.13702*, 2023.

Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let's verify step by step. In *International Conference on Learning Representations*, 2024.

Timothy P. Lillicrap, Jonathan J. Hunt, Alexander Pritzel, Nicolas Heess, Tom Erez, Yuval Tassa, David Silver, and Daan Wierstra. Continuous control with deep reinforcement learning. In *International Conference on Learning Representations*, 2016.

Andrew Kachites McCallum. *Reinforcement Learning with Selective Perception and Hidden State*. PhD thesis, University of Rochester, 1996.

Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Andrei A. Rusu, Joel Veness, Marc G. Bellemare, Alex Graves, Martin Riedmiller, Andreas K. Fidjeland, Georg Ostrovski, et al. Human-level control through deep reinforcement learning. *Nature*, 518(7540):529–533, 2015.

HyunJi Alex Nam, Scott Fleming, and Emma Brunskill. Reinforcement learning with state observation costs in action-contingent noiselessly observable Markov decision processes. In *Advances in Neural Information Processing Systems*, 2021.

Andrew Y. Ng, Daishi Harada, and Stuart Russell. Policy invariance under reward transformations: Theory and application to reward shaping. In *International Conference on Machine Learning*, 1999.

OpenAI. Learning to reason with LLMs. https://openai.com/index/learning-to-reason-with-llms/, 2024.

OpenAI. Evaluating chain-of-thought monitorability. https://openai.com/index/evaluating-chain-of-thought-monitorability/, 2025.

Alexander Pan, Kush Bhatia, and Jacob Steinhardt. The effects of reward misspecification: Mapping and mitigating misaligned models. In *International Conference on Learning Representations*, 2022.

Stuart Russell and Eric Wefald. Principles of metareasoning. *Artificial Intelligence*, 49(1–3): 361–395, 1991.

Tom Schaul, André Barreto, John Quan, and Georg Ostrovski. The phenomenon of policy churn. In *Advances in Neural Information Processing Systems*, 2022.

Amrith Setlur, Chirag Nagpal, Adam Fisch, Xinyang Geng, Jacob Eisenstein, Rishabh Agarwal, Alekh Agarwal, Jonathan Berant, and Aviral Kumar. Rewarding progress: Scaling automated process verifiers for LLM reasoning. In *International Conference on Learning Representations*, 2025.

Satinder P. Singh, Tommi Jaakkola, and Michael I. Jordan. Learning without state-estimation in partially observable Markovian decision processes. In *Proceedings of the Eleventh International Conference on Machine Learning*, pages 284–292, 1994.

Joar Skalse, Nikolaus H. R. Howe, Dmitrii Krasheninnikov, and David Krueger. Defining and characterizing reward hacking. In *Advances in Neural Information Processing Systems*, 2022.

Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. Scaling LLM test-time compute optimally can be more effective than scaling model parameters. *arXiv preprint arXiv:2408.03314*, 2024.

Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. Language models don't always say what they think: Unfaithful explanations in chain-of-thought prompting. In *Advances in Neural Information Processing Systems*, 2023.

Gladys Tyen, Hassan Mansoor, Victor Cărbune, Peter Chen, and Tony Mak. LLMs cannot find reasoning errors, but can correct them given the error location. In *Findings of the Association for Computational Linguistics: ACL 2024*, 2024.

Jonathan Uesato, Nate Kushman, Ramana Kumar, Francis Song, Noah Siegel, Lisa Wang, Antonia Creswell, Geoffrey Irving, and Irina Higgins. Solving math word problems with process- and outcome-based feedback. *arXiv preprint arXiv:2211.14275*, 2022.

Hado van Hasselt, Arthur Guez, and David Silver. Deep reinforcement learning with double Q-learning. In *AAAI Conference on Artificial Intelligence*, 2016.

Peiyi Wang, Lei Li, Zhihong Shao, R. X. Xu, Damai Dai, Yifei Li, Deli Chen, Yu Wu, and Zhifang Sui. Math-shepherd: Verify and reinforce LLMs step-by-step without human annotations. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024.

Steven D. Whitehead and Dana H. Ballard. Learning to perceive and act by trial and error. *Machine Learning*, 7(1):45–83, 1991.

Tian Xie, Zitian Gao, Qingnan Ren, Haoming Luo, Yuqian Hong, Bryan Dai, Joey Zhou, Kai Qiu, Zhirong Wu, and Chong Luo. Logic-RL: Unleashing LLM reasoning with rule-based reinforcement learning. *arXiv preprint arXiv:2502.14768*, 2025.

A Environment Details

Transitions. Given the current true value v, an operation type is drawn uniformly from those available, and then its operand uniformly: `add` k with $k \in \{1, \ldots, 4\}$; `sub` k with $k \in \{1, \ldots, \min(4, v-1)\}$ if $v > 2$; and `mul` k with $k \in \{2, 3\}$ if $|v| \leq 16$. The corrupted version of `add`/`sub` k uses k+1, and that of `mul` k is `add` k. If the corrupted operation happens to give the same value, it is replaced by `add` k+2. A corrupted step therefore always changes the value (unit-tested).

Fault prior. Conditional on a fault, its position $h \in \{0, \ldots, d-1\}$ has weight $1 + 5\,(1 - |h - c| / \max(1, c))$ with $c = (d-1)/2$, i.e. it peaks in the middle of the chain.

Cue selectivity metric. At every decision point where the current step $t > 0$ has not been checked and `VERIFY` is available, we record the cue c_t and whether the agent chose `VERIFY`. $P(\texttt{VERIFY} \mid c)$ pools these counts over all evaluation episodes of a seed.

Validation against the original implementation. The environment was rewritten for this version. `make equivalence` (`scripts/check_legacy_equivalence.py`) compares it with the original implementation, vendored from commit `81ec8cb` into `scripts/legacy/` with formatting changes only, by playing identical random action sequences in both. Over 86,112 steps from 18,000 episodes, covering all reward schemes and both budget-penalty settings, the two agree exactly on transitions, masks, rewards, termination and absolute observations. The only difference is the observation returned after the terminal `COMMIT`, which no learner uses.

B Hyperparameters and Compute

Table 8: Default environment and learner settings.

Setting	Value
Training / held-out lengths	3–6 / 7–9 and 10–14
Fault probability; cue hit / false-positive rate	0.8; 0.85 / 0.15
Budget; action costs (`THINK`, `VERIFY`, `REVISE`)	$d + 6$; $(1, 1, 1)$
Rewards: success / failure / per-unit cost	$+1$ / -1.2 / -0.04
Monitorability-aware coefficients (Eq. 1)	$0.02, 0.25, 0.18, 0.02, 0.06, 0.30$
Network	MLP, 2 hidden layers of 64 ReLU units
Optimiser; learning rate; gradient clip	Adam; 10^{-3}; 1.0 (global norm)
Discount; loss	0.99; Huber
Replay size; batch size; warm-up	25,000; 64; 500 steps
Target network	hard copy every 200 environment steps
Exploration	ϵ: $0.5 \to 0.02$ linearly over 85% of training
Training episodes per run	20,000
Seeds	10 per condition (5 in the sensitivity study)
Validation for selection	200 episodes of lengths 3–6 every 1,000 episodes
Evaluation	500 episodes per seed and setting

A training run takes about four minutes on one CPU core. The main experiment (60 runs), the ablations (80 runs; the full-reward condition is shared with the main experiment), the sensitivity study (70 runs, ten of them twice as long), the follow-ups (130 runs), the encoder control (10 runs) and the step-bonus sweep (60 runs) are 410 distinct runs, 32 CPU-hours

in total (4 early runs have no recorded time and are counted at the mean of runs of the same length), about 8 hours on a 4-core machine. No GPU is used. Training is deterministic on CPU given a seed; `make reproduce` retrains everything into a fresh directory (420 runs, since the shared condition is retrained there) and checks every checkpoint bitwise against the saved ones. Bitwise identity requires the pinned versions in `requirements-lock.txt` and the same CPU instruction set: with AVX2 or without vector extensions torch's kernels round differently, which changes every training trajectory. The platform (Python, torch and numpy versions, CPU capability, BLAS) is recorded in `run_info.json` from the step-bonus sweep on (x86-64 Linux, AVX-512, MKL); the other experiments predate that record. `make spot-check` (about 5 minutes on 6 cores, 9 on 4) retrains one run from each experiment and compares it bitwise; on an AVX-512 machine all six reproduce exactly. On other hardware, `scripts/compare_summaries.py` compares retrained and saved success per method, paired by seed, and checks whether the headline comparisons keep their sign and resolution.

Hyperparameter pilot. Table 9 lists the two-seed pilot behind the choice of learner settings (§4). It was run with an earlier revision of this code and is archived, not regenerated.

Table 9: Hyperparameter pilot (2 seeds, selected checkpoints). The setting used throughout is learning rate 10^{-3} with a hard target update every 200 steps.

Reward	Obs.	Learning rate	Target update	Success 3–6	Success 7–9
Monitorability-aware	abs	10^{-3}	hard, 200	0.64	0.29
Monitorability-aware	abs	10^{-3}	hard, 1000	0.66	0.24
Monitorability-aware	abs	$3{\cdot}10^{-4}$	hard, 200	0.23	0.01
Monitorability-aware	abs	$3{\cdot}10^{-4}$	hard, 1000	0.44	0.27
Monitorability-aware	inv	10^{-3}	hard, 200	0.88	0.85
Monitorability-aware	inv	10^{-3}	hard, 1000	0.39	0.27
Monitorability-aware	inv	$3{\cdot}10^{-4}$	hard, 200	0.77	0.68
Monitorability-aware	inv	$3{\cdot}10^{-4}$	hard, 1000	0.48	0.43
Monitorability-aware	inv	10^{-3}	Polyak τ=0.005	0.89	0.87
Monitorability-aware	inv	$3{\cdot}10^{-4}$	Polyak τ=0.005	0.89	0.82
Monitorability-aware	abs	10^{-3}	Polyak τ=0.005	0.74	0.29
Monitorability-aware	abs	$3{\cdot}10^{-4}$	Polyak τ=0.005	0.52	0.15
Outcome-only	inv	10^{-3}	Polyak τ=0.005	0.79	0.75
Outcome-only	inv	$3{\cdot}10^{-4}$	Polyak τ=0.005	0.40	0.31

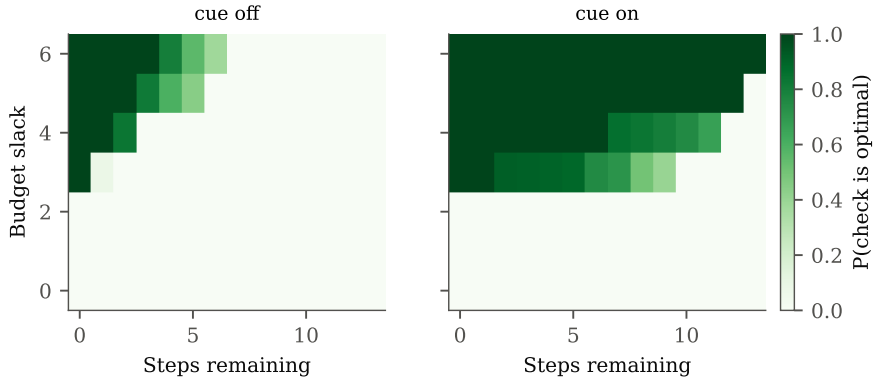


Figure 5: Where the exact planner checks the step it has just executed, by steps remaining and budget slack, for un-cued and cued steps. Colour is the fraction of chain lengths 3–14 for which checking is optimal; the fault prior peaks mid-chain, so the decision also depends on position. At fixed slack and cue the decision changes with the number of remaining steps, which a length-invariant observation does not reveal.

C Last-Checkpoint Results

Table 10: As Table 2, but evaluating the last checkpoint of each run instead of the one selected on training-length validation. Heuristics are repeated for reference.

Method	Success 3–6	Success 7–9	Success 10–14	Detect 7–9	Cost-adj. 7–9
Random	0.00 [0.00, 0.01]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.02 [0.01, 0.02]	−0.09 [−0.09, −0.08]
Always THINK	0.20 [0.19, 0.21]	0.20 [0.19, 0.22]	0.20 [0.18, 0.21]	0.00 [0.00, 0.00]	−0.20 [−0.21, −0.18]
Verify every step	0.80 [0.79, 0.81]	0.06 [0.06, 0.07]	0.00 [0.00, 0.00]	0.93 [0.91, 0.94]	−0.63 [−0.64, −0.63]
Cue-triggered	0.87 [0.86, 0.89]	0.88 [0.87, 0.90]	0.85 [0.84, 0.86]	0.85 [0.84, 0.87]	0.33 [0.31, 0.34]
Budget-aware	0.97 [0.96, 0.97]	0.82 [0.81, 0.84]	0.68 [0.66, 0.70]	0.78 [0.76, 0.80]	0.19 [0.17, 0.20]
Planner (no back-checking)	0.99 [0.99, 0.99]	0.94 [0.93, 0.94]	0.90 [0.89, 0.92]	0.92 [0.91, 0.93]	0.33 [0.32, 0.34]
Outcome (inv)	0.40 [0.27, 0.52]	0.33 [0.21, 0.46]	0.30 [0.19, 0.41]	0.25 [0.04, 0.45]	−0.31 [−0.43, −0.18]
Naive process (inv)	0.41 [0.25, 0.58]	0.36 [0.21, 0.51]	0.31 [0.20, 0.43]	0.58 [0.42, 0.75]	−0.30 [−0.45, −0.14]
Monitorability (inv)	0.77 [0.68, 0.86]	0.63 [0.51, 0.76]	0.52 [0.41, 0.64]	0.81 [0.74, 0.89]	−0.02 [−0.15, 0.10]
Outcome (abs)	0.24 [0.22, 0.26]	0.12 [0.05, 0.19]	0.08 [0.02, 0.14]	0.06 [0.00, 0.16]	−0.20 [−0.34, −0.06]
Naive process (abs)	0.23 [0.17, 0.30]	0.12 [0.05, 0.20]	0.05 [0.00, 0.11]	0.19 [0.02, 0.35]	−0.28 [−0.41, −0.15]
Monitorability (abs)	0.63 [0.59, 0.68]	0.24 [0.15, 0.34]	0.03 [0.00, 0.07]	0.51 [0.36, 0.65]	−0.34 [−0.43, −0.24]

Table 11: Reward ablations evaluated at the last checkpoint instead of the selected one (cf. Table 5).

Variant	Success 3–6	Success 7–9	Success 10–14	$P(V \mid c{=}1)$	$P(V \mid c{=}0)$
Full reward	0.77 [0.68, 0.86]	0.63 [0.51, 0.76]	0.52 [0.41, 0.64]	0.88	0.14
− cue bonus/penalty	0.63 [0.46, 0.79]	0.45 [0.29, 0.61]	0.37 [0.24, 0.50]	0.62	0.22
− unchecked-gap term	0.65 [0.53, 0.77]	0.53 [0.39, 0.68]	0.50 [0.37, 0.63]	0.86	0.07
− wasted-check penalty	0.55 [0.35, 0.75]	0.48 [0.30, 0.66]	0.43 [0.27, 0.58]	0.84	0.09
− safe/fault check bonuses	0.78 [0.69, 0.86]	0.59 [0.49, 0.70]	0.45 [0.32, 0.58]	0.66	0.11
− cue, gap and waste terms	0.62 [0.46, 0.77]	0.51 [0.38, 0.64]	0.41 [0.31, 0.51]	0.70	0.18
cue bonus only	0.63 [0.47, 0.79]	0.54 [0.40, 0.69]	0.45 [0.32, 0.58]	0.79	0.14
− per-action cost	0.52 [0.34, 0.71]	0.44 [0.29, 0.59]	0.37 [0.25, 0.48]	0.89	0.09
REVISE disabled	0.20 [0.19, 0.21]	0.18 [0.16, 0.20]	0.15 [0.12, 0.19]	0.63	0.11

D Behavioural Diagnostics

Table 12: Behaviour at held-out lengths 7–9 (mean over 10 seeds, selected checkpoints). Non-repair `REVISE`s do not follow an UNSAFE check and cannot repair anything; "success | detect" is the fraction of episodes with a caught fault that end in success; "early commit" is the fraction of episodes committed before the chain is finished although the budget could still finish it.

Method	Checks/ep.	Non-repair REVISE/ep.	Detect	Success | detect	Early commit	Commit at $t{=}0$
Cue-triggered	1.76	0.00	0.85	1.00	0.00	0.00
Budget-aware	3.52	0.00	0.78	1.00	0.00	0.00
Planner (no back-checking)	2.66	0.00	0.92	1.00	0.00	0.00
Outcome (inv)	2.23	0.83	0.61	0.84	0.00	0.00
Naive process (inv)	3.38	0.42	0.76	0.91	0.00	0.00
Monitorability (inv)	2.70	0.76	0.85	0.97	0.00	0.00
Outcome (abs)	1.30	1.04	0.09	0.58	0.33	0.30
Naive process (abs)	2.15	0.78	0.30	0.22	0.30	0.14
Monitorability (abs)	2.01	1.15	0.60	0.35	0.18	0.07

Table 14: Success under shifts at held-out lengths 7–9, without retraining (mean over 10 seeds ± half-width of the 95% t-interval). The planner is recomputed for each shifted environment. p_f: fault probability (train: 0.8); c_V: verification cost (train: 1); slack: extra budget beyond the chain length (train: 6); clean/noisy cue: hit/false-positive rates 0.95/0.05 and 0.60/0.35 (train: 0.85/0.15).

Method	default	p_f=0.6	p_f=0.9	c_V=2	c_V=3	slack 4	slack 8	clean cue	noisy cue
Always THINK	0.20 ±0.01	0.41 ±0.02	0.10 ±0.01	0.20 ±0.01	0.19 ±0.01	0.20 ±0.01	0.21 ±0.02	0.21 ±0.02	0.20 ±0.02
Cue-triggered	0.88 ±0.01	0.91 ±0.01	0.86 ±0.01	0.75 ±0.01	0.49 ±0.01	0.73 ±0.01	0.88 ±0.02	0.96 ±0.01	0.62 ±0.01
Budget-aware	0.82 ±0.02	0.88 ±0.01	0.81 ±0.01	0.66 ±0.01	0.22 ±0.01	0.67 ±0.01	0.95 ±0.01	0.93 ±0.01	0.65 ±0.01
Planner (no back-checking)	0.94 ±0.01	0.96 ±0.01	0.93 ±0.01	0.82 ±0.01	0.65 ±0.01	0.82 ±0.01	0.98 ±0.00	0.98 ±0.00	0.80 ±0.01
Outcome (inv)	0.62 ±0.14	0.72 ±0.11	0.56 ±0.16	0.51 ±0.14	0.22 ±0.11	0.42 ±0.15	0.41 ±0.17	0.75 ±0.17	0.41 ±0.09
Naive process (inv)	0.76 ±0.11	0.83 ±0.08	0.74 ±0.12	0.56 ±0.15	0.26 ±0.14	0.44 ±0.14	0.68 ±0.12	0.87 ±0.12	0.55 ±0.08
Monitorability (abs)	0.31 ±0.11	0.41 ±0.10	0.27 ±0.11	0.25 ±0.09	0.11 ±0.04	0.23 ±0.10	0.36 ±0.11	0.37 ±0.12	0.22 ±0.07
Monitorability (inv)	0.86 ±0.03	0.90 ±0.02	0.85 ±0.04	0.69 ±0.07	0.24 ±0.11	0.49 ±0.13	0.65 ±0.18	0.94 ±0.03	0.64 ±0.04

Table 13: As Table 12, at lengths 10–14.

Method	Checks/ep.	Non-repair REVISE/ep.	Detect	Success | detect	Early commit	Commit at t=0
Cue-triggered	2.37	0.00	0.86	0.95	0.00	0.00
Budget-aware	3.74	0.00	0.60	1.00	0.00	0.00
Planner (no back-checking)	2.73	0.00	0.88	1.00	0.00	0.00
Outcome (inv)	2.51	0.85	0.53	0.79	0.00	0.00
Naive process (inv)	3.69	0.42	0.67	0.85	0.00	0.00
Monitorability (inv)	3.19	0.74	0.75	0.91	0.00	0.00
Outcome (abs)	0.39	0.79	0.02	0.42	0.74	0.72
Naive process (abs)	0.47	0.22	0.07	0.05	0.81	0.62
Monitorability (abs)	1.08	1.06	0.15	0.01	0.70	0.56

E Robustness to Environment Shifts

Table 14 evaluates policies at lengths 7–9 under shifts they were not trained on.

- **Fault rate.** The invariant monitorability-aware agent is robust to the fault rate (0.90 at p_f=0.6, 0.85 at p_f=0.9).
- **Cue quality.** It tracks the cue-triggered heuristic when cue quality changes (clean: 0.94 vs. 0.96; noisy: 0.64 vs. 0.62). This confirms that the policy relies on the cue rather than on a fixed checking schedule.
- **Verification cost.** It degrades when checks cost more than in training (0.24 at c_V=3, against 0.49 for the cue-triggered heuristic and 0.65 for the planner), because it keeps checking at a rate the budget no longer supports.
- **Budget slack.** It is sensitive to the slack in *both* directions. With slack 4, success falls to 0.49 (cue-triggered: 0.73; planner: 0.82), although every slack value then stays inside the training range. This is the allocation problem of §6.5 again: a checking rate tuned to slack 6 runs out of budget. With slack 8, success falls to 0.65. Here the budget-aware heuristic improves (0.95), and the cue-triggered heuristic, which never spends extra slack, is unchanged (0.88). With slack 8 the slack feature also takes values never seen in training. The "invariant" encoder is invariant to length, not to budget.

F Sensitivity to the Learner's Hyperparameters

The main experiment uses the earlier version's DQN settings. To check that the encoder effect and the reward comparison are not specific to them, we retrain the monitorability-aware agents with both encoders, and outcome-only and naive agents with the invariant encoder, under three alternatives, five seeds each: learning rate $3{\cdot}10^{-4}$; hard target updates every 1,000 steps; and Polyak-averaged targets ($\tau = 0.005$) [Lillicrap et al., 2016] (Table 16). None of them removes the churn. In every setting the encoder gap reproduces: on lengths 10–14 the invariant encoder beats the absolute one by +0.64, +0.61 and +0.71 respectively, in 5/5, 5/5 and 5/5 seeds. With five seeds, the smallest attainable sign-flip p-value is 0.0625.

Table 15: The reward comparison (invariant observations) under each learner setting: the fraction of late checkpoints (after 5,000 episodes) that collapse to training-length success ≤ 0.35, and selected-checkpoint success at lengths 7–9 with its 95% t-interval. Default: 10 seeds; alternatives: 5 seeds.

Learner	Collapsed: outcome	naive	MA	Success 7–9: outcome	naive	MA
default (main experiment)	85%	62%	21%	0.62 [0.48, 0.75]	0.76 [0.64, 0.87]	0.86 [0.83, 0.89]
lr $3{\cdot}10^{-4}$	81%	49%	19%	0.48 [0.29, 0.68]	0.64 [0.43, 0.84]	0.82 [0.75, 0.89]
hard sync 1000	96%	75%	17%	0.40 [0.18, 0.62]	0.58 [0.41, 0.75]	0.82 [0.74, 0.91]
Polyak τ=0.005	87%	80%	19%	0.52 [0.18, 0.86]	0.36 [0.26, 0.46]	0.86 [0.84, 0.88]
40k episodes	83%	–	22%	0.53 [0.32, 0.74]	–	0.83 [0.75, 0.90]

Table 16: Sensitivity of the encoder comparison to DQN hyperparameters (monitorability-aware reward; mean over 5 seeds, 95% t-interval). Main-experiment values with the default settings are in Table 2.

Variant	Success 3–6	Success 7–9	Success 10–14	$P(V \mid c{=}1)$	$P(V \mid c{=}0)$
abs, lr $3{\cdot}10^{-4}$	0.72 [0.58, 0.87]	0.24 [0.05, 0.42]	0.03 [0.00, 0.08]	0.76	0.11
inv, lr $3{\cdot}10^{-4}$	0.88 [0.85, 0.91]	0.82 [0.75, 0.89]	0.67 [0.52, 0.82]	0.95	0.11
abs, hard sync 1000	0.80 [0.77, 0.83]	0.29 [0.16, 0.42]	0.02 [0.00, 0.04]	0.70	0.20
inv, hard sync 1000	0.90 [0.88, 0.91]	0.82 [0.74, 0.91]	0.63 [0.50, 0.76]	0.95	0.18
abs, Polyak τ=0.005	0.76 [0.73, 0.79]	0.26 [0.15, 0.37]	0.04 [0.00, 0.10]	0.66	0.18
inv, Polyak τ=0.005	0.90 [0.88, 0.93]	0.86 [0.84, 0.88]	0.75 [0.67, 0.83]	0.83	0.13
inv, 40k episodes	0.90 [0.88, 0.92]	0.83 [0.75, 0.90]	0.69 [0.51, 0.88]	0.86	0.13

The reward comparison also holds (Table 15), including with twice the training (40,000 episodes, with the exploration schedule stretched to match; outcome-only and monitorability-aware agents only). In every setting, late checkpoints of the monitorability-aware agents collapse far less often (19%, 17% and 19%) than those of outcome-only agents (81%, 96% and 87%). The monitorability-aware agents beat outcome-only at lengths 7–9 by +0.34, +0.42 and +0.34, in 5/5, 5/5 and 4/5 seeds. Naive process reward again collapses less often than outcome-only (49%, 75% and 80%), but whether it does better on held-out lengths depends on the learner: at 7–9 it is ahead by 0.15 with the lower learning rate (5/5 seeds) and behind with Polyak averaging (a difference of −0.16, 2/5 seeds ahead).

G Additional Follow-up Results

Memory of a caught fault. Adding one bit that records whether an UNSAFE check has been seen removes the aliasing between repairs and no-ops. Wasted `REVISE`s at 10–14 fall from 0.74 to 0.23 per episode. Training-length success rises by +0.03 (9/10 seeds, $p = 0.004$). Held-out success at selected checkpoints does not change resolvably ($p = 0.996$ at 10–14), because the agent spends the recovered slack on more un-cued checks (0.22 vs. 0.12). The allocation explanation of §6.5 is therefore only partly confirmed: removing one source of waste was not enough. Checkpoint collapse is lower with the memory bit (10% vs. 21%), but the intervals overlap. At the last checkpoint the point estimates favour the bit by +0.17 at 7–9 and +0.17 at 10–14, similar in size to the repair payment's gain below, but 10 seeds resolve neither ($p = 0.074$ and 0.080). The bit is not a test of the aliasing explanation of §6.2: it stays set after a repair, so after continuing past a caught fault the agent still sees observations that also occur on track (18% of them, against 96% without the bit). The repair-pending bit of §6.9 is that test.

Repair payment: selected and last checkpoints. The benefit shows up in *reliability* rather than peak performance. At selected checkpoints the repair-paid reward does not beat the original, because selection already discards collapsed checkpoints. It may even be slightly worse there: −0.07 [−0.15, 0.01] at 7–9 ($p = 0.080$) and −0.10 [−0.24, 0.04] at 10–14, differences that 10 seeds do not resolve. At the last checkpoint it is better: +0.12 [0.03, 0.21] at 7–9 ($p = 0.016$) and +0.12 at 10–14 ($p = 0.049$).

The two last-checkpoint agents behave differently. The repair-paid agent checks less (1.81 vs. 2.57 checks per episode) and catches fewer faults (0.70 vs. 0.81). But it almost always completes the repairs it starts (0.98 vs. 0.69).

Slack per remaining step. Adding slack per remaining step, $\mathrm{clip}(\mathrm{slack}/(d-t), 0, 2)/2$, gives the agent the information that §6.5 identifies as missing. Its values stay inside the interval seen in training, but many of them, and most of the observation vectors they form, never occur on training lengths. Observations at 51% of decision points at lengths 7–9 and 72% at 10–14 never appear in the training-length reference sample, against 2% and 5% for the invariant encoder. Held-out success falls to 0.46 at 7–9 and 0.28 at 10–14 (-0.46, $p = 0.002$). Most of the loss comes from the repair decision rather than from allocation. (Half of these runs select a checkpoint from the first 5,000 episodes: median selected episode 7,000, 5/10 within the burn-in; the numbers below partly describe immature policies.) Fault detection stays high (0.81), but repairs after a detection are completed in only 0.39 of episodes at 7–9 and 0.15 at 10–14, against 0.97 and 0.91 for the reference. The extra feature destabilises the same fragile choice after an UNSAFE check that §6.2 studies, and it also costs -0.07 on training lengths. This run therefore does not show whether remaining-length information would help allocate the budget; it shows that adding it this way made the repair decision worse.

Checkpoint selection with the repair-pending bit. At selected checkpoints the bit raises training-length success under every reward but lowers held-out success on long chains for the monitorability-aware agents (-0.14 [-0.26, -0.01] at 10–14) and, in point estimate, the naive ones (-0.10 [-0.25, 0.05]). For the monitorability-aware agents this is most likely a selection effect rather than a cost of the bit (for the naive agents it is not: their last checkpoints are no better). Of these selected checkpoints, 2/10 come from the first 5,000 episodes. With the bit, training-length validation success saturates (median selected value 0.96, against 0.91), so selection chooses among checkpoints that differ mainly in how they behave on long chains, and it tends to pick ones that check many un-cued steps (0.39 of un-cued decision points at 10–14, against 0.12) and run short of slack. The last checkpoints of the same runs check un-cued steps less often (0.17) and reach 0.83 [0.76, 0.89] at 10–14, an unresolved difference from the cue-triggered heuristic of -0.02 [-0.08, 0.04] (a comparison we made after seeing the long-chain results). Training-length selection is a poor guide once training-length performance saturates; it also flatters the reference agent's selected numbers. Compared like for like, last checkpoint against last checkpoint, the bit helps by $+0.26$ at 7–9 and $+0.30$ at 10–14.

H Value Probe After a Caught Fault

Table 17: Value probe (`scripts/q_probe.py`) at decision points right after an UNSAFE result where the repair is still affordable, training lengths, selected checkpoints unless noted (means over points, then over 10 seeds). R/T: `REVISE`/`THINK`. "True" values are discounted returns of playing each action and then the same greedy policy, computed exactly from a copy of the environment. "Error": learned minus true value. "Offset": the same error for the greedy action at the first decision point of an episode, where the hidden state follows the task prior and no history can create aliasing. "Excess error": error minus offset. "Detect, no repair": fraction of late checkpoints that collapse while still catching faults. Seeds without any such decision point are left out of a mean (the last outcome-only checkpoints have them in only 5 of 10 seeds).

Agent	Detect, no repair	THINK after UNSAFE (sel. / last)	Error R / T	Offset	Excess error R / T	Excess, last ckpt. R / T
Outcome-only	15%	5% / 22%	−1.38 / +0.64	−0.76	−0.62 / +1.40	−0.96 / +0.88
Naive process	29%	2% / 28%	−1.36 / +0.72	−1.00	−0.36 / +1.72	−1.21 / +0.90
Monitorability-aware (MA)	21%	0% / 25%	−0.58 / +1.09	−0.45	−0.13 / +1.54	−0.42 / +1.15
MA + memory bit	7%	0% / 9%	−0.05 / +0.59	0.00	−0.05 / +0.59	−0.23 / +0.77
MA, bonus paid on repair	1%	1% / 0%	−0.86 / +1.40	−0.54	−0.32 / +1.94	−0.42 / +1.83
MA + memory bit, bonus on repair	0%	1% / 3%	−0.11 / +2.02	−0.04	−0.07 / +2.06	−0.13 / +2.01
Outcome-only + pending bit	0%	7% / 0%	−0.91 / +0.31	−0.71	−0.20 / +1.03	−0.78 / +0.98
Naive process + pending bit	0%	0% / 1%	−0.70 / +1.11	−0.49	−0.20 / +1.61	−0.65 / +1.29
MA + pending bit	0%	0% / 0%	−0.24 / +0.93	−0.14	−0.10 / +1.07	−0.15 / +1.23
Outcome-only + last 8 actions	0%	1% / 0%	+0.04 / +0.15	−0.01	+0.05 / +0.16	−0.16 / +0.20
Naive process + last 8 actions	0%	0% / 1%	+0.04 / +0.26	0.02	+0.02 / +0.24	−0.13 / +0.24
MA + last 8 actions	0%	0% / 0%	0.00 / +0.44	0.04	−0.05 / +0.40	−0.07 / +0.47
Outcome-only + last 3 actions	0%	0% / 0%	−0.33 / +1.22	−0.25	−0.08 / +1.47	−0.25 / +1.29
Naive process + last 3 actions	0%	0% / 0%	−0.18 / +1.20	−0.14	−0.04 / +1.35	−0.18 / +1.54
MA + last 3 actions	0%	0% / 1%	−0.08 / +1.30	−0.06	−0.01 / +1.37	−0.11 / +1.28

I Changes from the Earlier Version

The first public version of this repository (commit `81ec8cb`) reported that monitorability-aware RL reached 0.60 success on training lengths but only 0.11 on lengths 7–9, and concluded that the learned checking behaviour was brittle under length shift. This version changes the following.

1. **Representation control.** The original observation contained absolute position and depth features. We add a length-invariant encoder and compare the two directly (§6.3). The encoder, not the reward, explains most of the length-generalization failure.
2. **Training length and checkpoint selection.** Runs were trained for 5,000 episodes, while learning curves were still rising, and evaluated at the last checkpoint. We train for 20,000 episodes, select checkpoints on training-length validation only, and report last-checkpoint results as well (Appendix C).
3. **Seed overlap.** Training episode seeds were $\texttt{seed}\cdot 10^5 + i$ and evaluation seeds $\texttt{seed}\cdot 10^6 + i$, which coincide for seed 0: 299 of seed 0's 300 training-length evaluation episodes were exact copies of training episodes. Because policies observe only depth, position, budget, cue and check results, an episode is determined (for a deterministic policy) by its depth, fault position and cue vector, and on training lengths every such configuration recurs many times in training anyway. The overlap therefore could not bias the reported numbers; we fixed it for hygiene by using independent random-number streams, and tests check that evaluation draws from its own stream.
4. **Localization metric.** The original "failed-episode localization score" gave full credit when the fault was on the last step and the agent committed without checking anything, and partial credit for any commit at or after the fault step; as a result the never-check policy scored 0.07. It inflated every method's diagnostic score without changing the main comparison. We replace it with fault detection, the probability that a check caught the fault, and add repair completion.
5. **Stronger baselines and statistics.** We add a budget-aware heuristic tuned on training lengths, use 10 seeds instead of 5, and report t-intervals, paired differences and permutation tests instead of standard errors.

6. **Learner details.** The replay buffer now samples minibatches with replacement (the original sampled without replacement), and the original's per-step `invalid_action_penalty` was dropped because masked agents never take invalid actions. Neither affects the environment's dynamics or rewards (Appendix A).
7. **Heuristic definitions.** The original heuristics committed as soon as the last step was executed, so they never checked the final step, and the cue-triggered heuristic re-checked a step after repairing it. Ours check the final step and never re-check. On the same evaluation episodes this raises cue-triggered success at lengths 3–6 from 0.82 to 0.87 (almost entirely from the final-step check) and verify-every-step success from 0.57 to 0.80. Learned agents are unaffected.